

STAT 626 HW 6

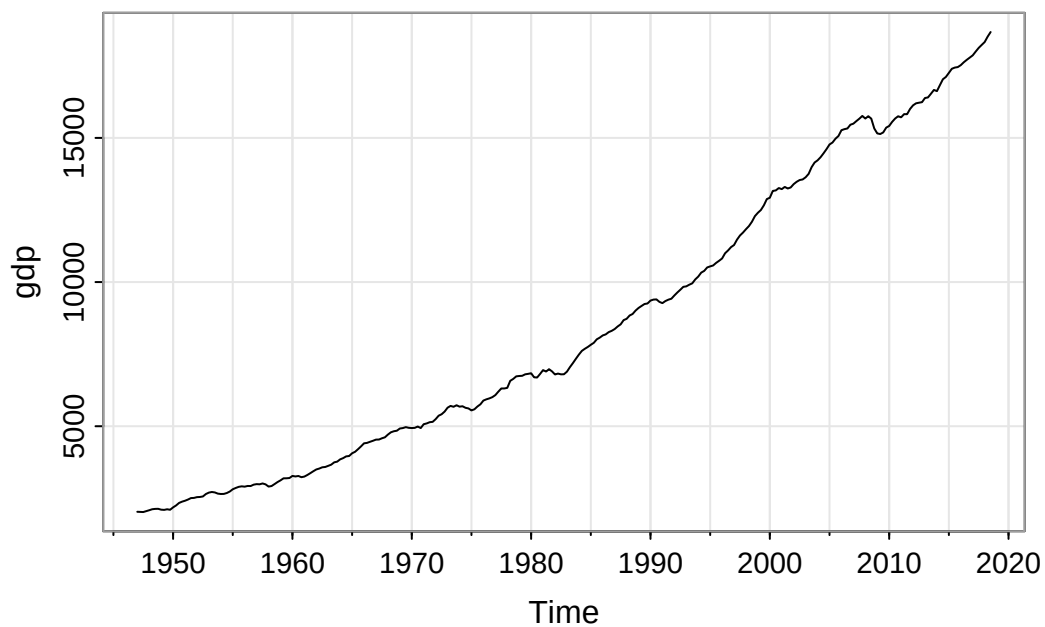
Jack Cunningham

```
library(astsa)
knitr::opts_chunk$set(echo = F)
knitr::opts_chunk$set(warning = F)
```

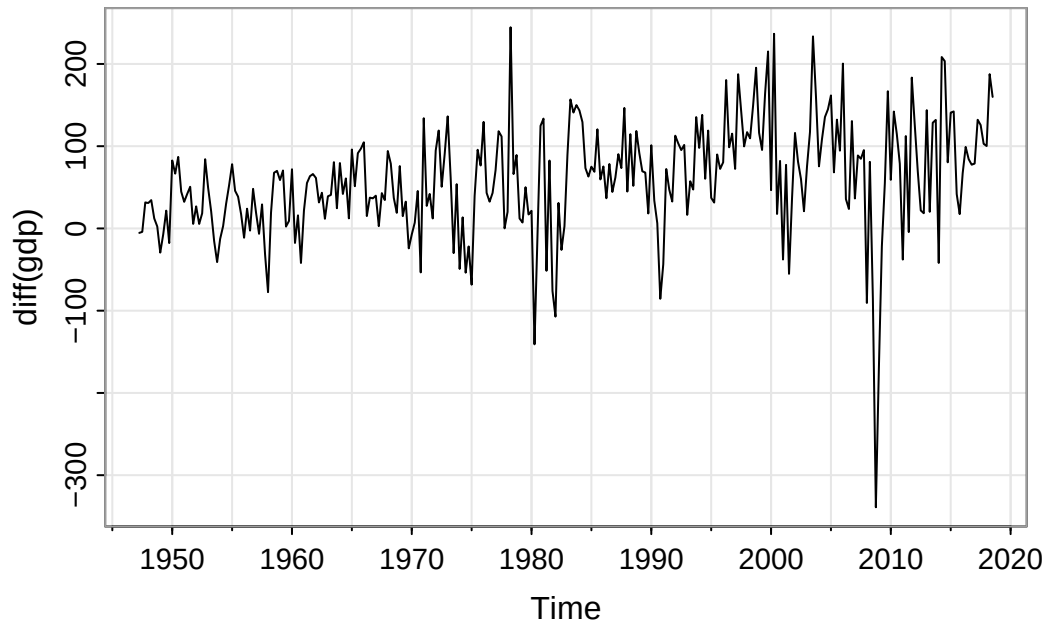
I.

5.2)

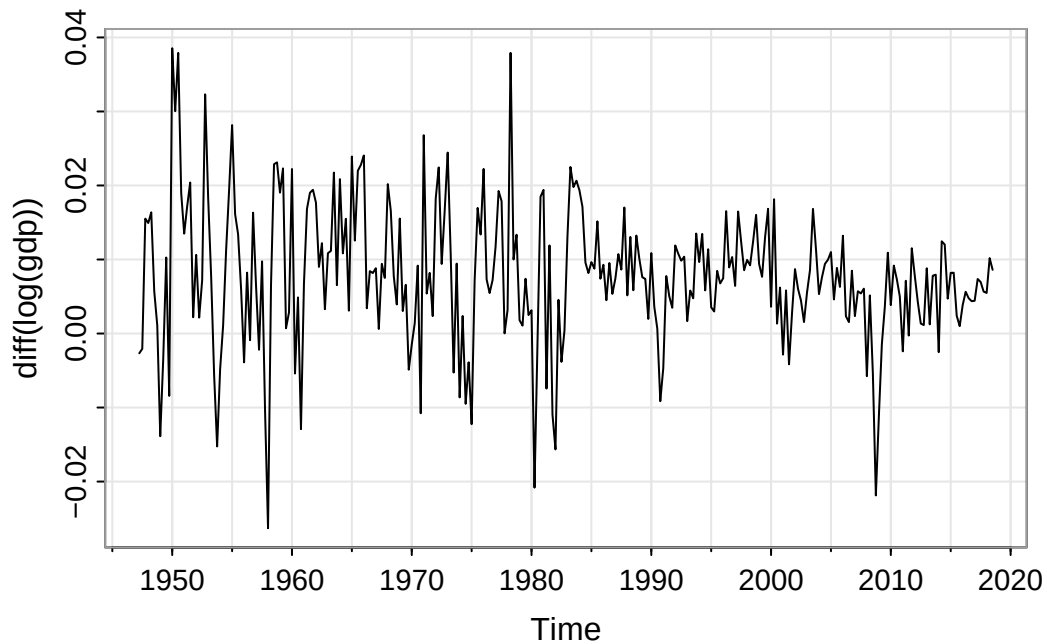
We start with a time series plot of GNP data:



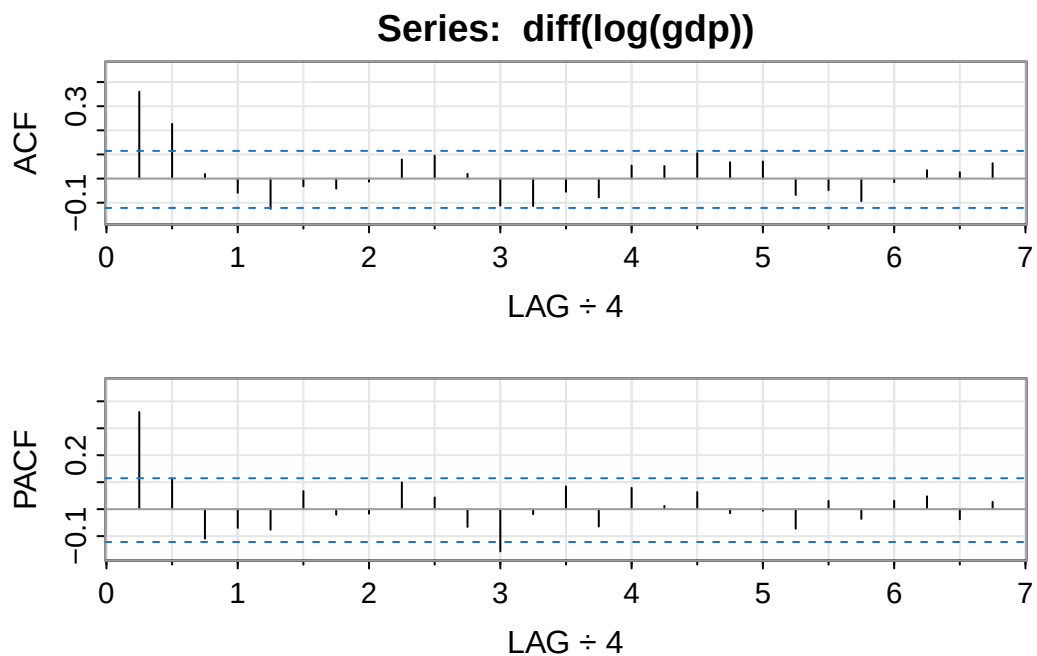
We see an upward trend over time, the first remedy would be to difference the data. We do so and plot again:



There is evidence of non-constant variance, as time passes the variance of the series increases. To remedy this we instead use $\log(x_t) - \log(x_{t-1})$, since we expect GNP to have stable and small percent changes:



The data is now close to stationarity. So we plot the ACF and PACF to identify the dependence orders of the model.



[,1] [,2] [,3] [,4] [,5] [,6] [,7] [,8] [,9] [,10] [,11] [,12] [,13]

[illegible]

The ACF drops to zero after two lags, PACF seems to tail off. My first model would therefore be a $ARIMA(0, 1, 2)$ model.

We now fit the model:

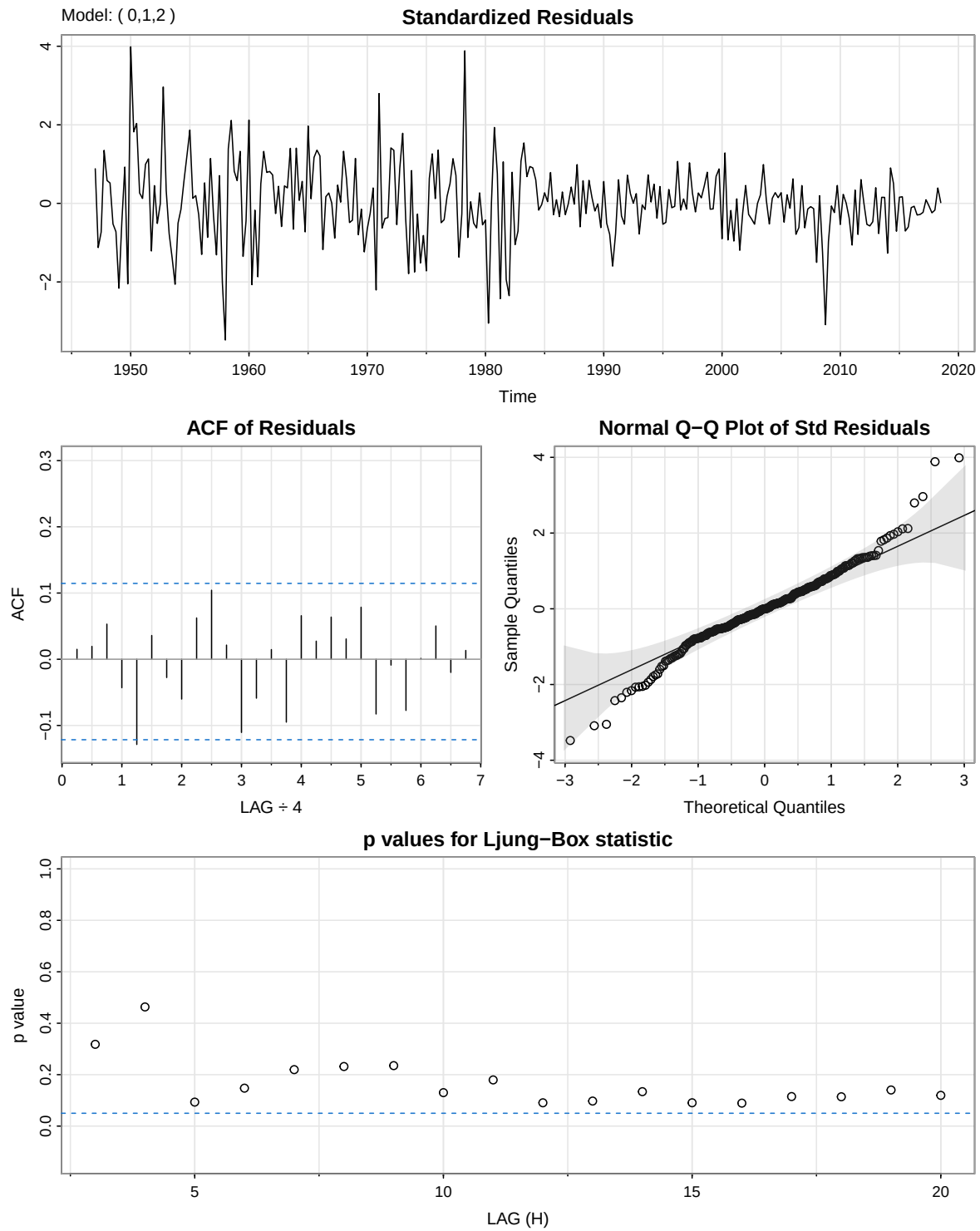
[illegible]

Coefficients:

	Estimate	SE	t.value	p.value
ma1	0.3070	0.0579	5.2987	0
ma2	0.2258	0.0547	4.1270	0
constant	0.0077	0.0008	9.8631	0

sigma^2 estimated as 7.464625e-05 on 283 degrees of freedom

AIC = -6.636309 AICc = -6.636011 BIC = -6.585176



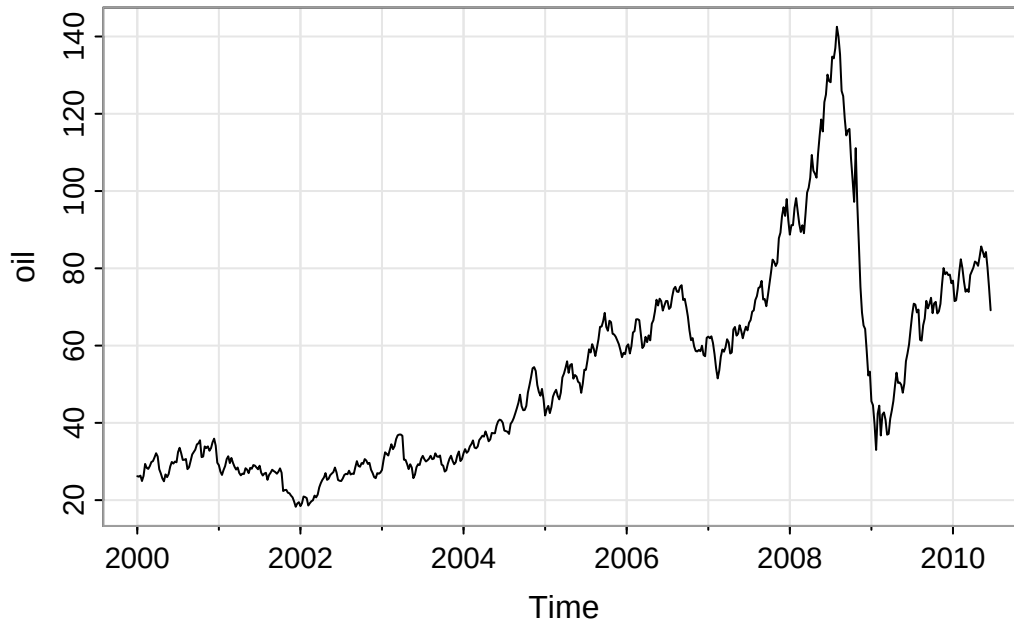
The residual plot looks pretty good, there does not appear to be correlation in the residuals

according to the ACF plot or the Ljung-Box plot. Residuals appear close to normal with slight deviations in the tails. Standardized residuals seem to point towards some evidence of non-constant variance, but overall are close to stationarity. Non-constant variance could likely be remedied by a GARCH model but that falls outside the bounds of this question.

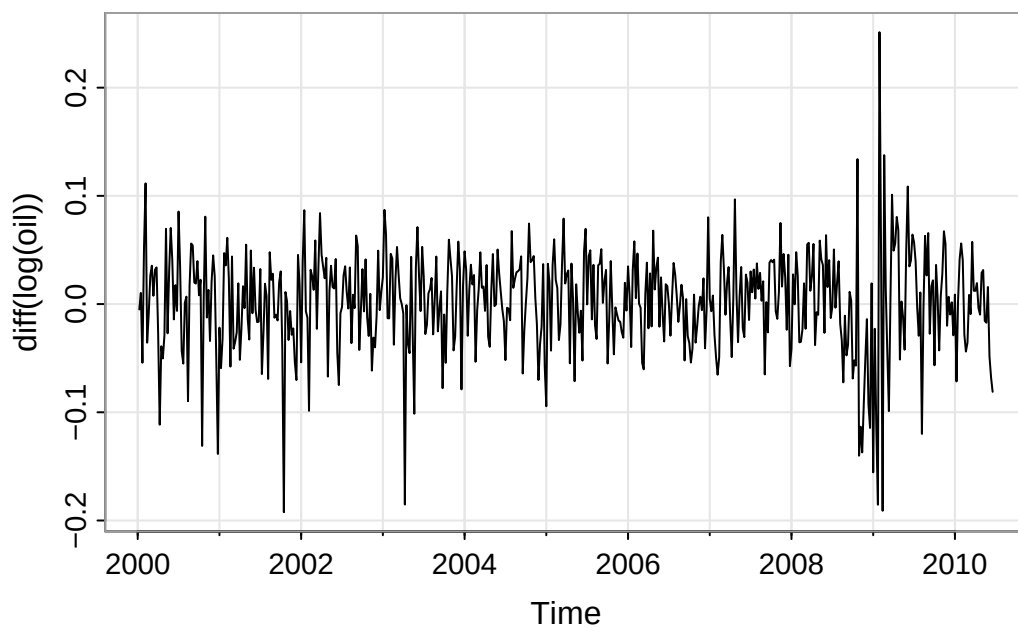
I would select an $ARIMA(0, 1, 2)$ model with a log transformation.

5.3)

We start with a time series plot of the crude oil prices:

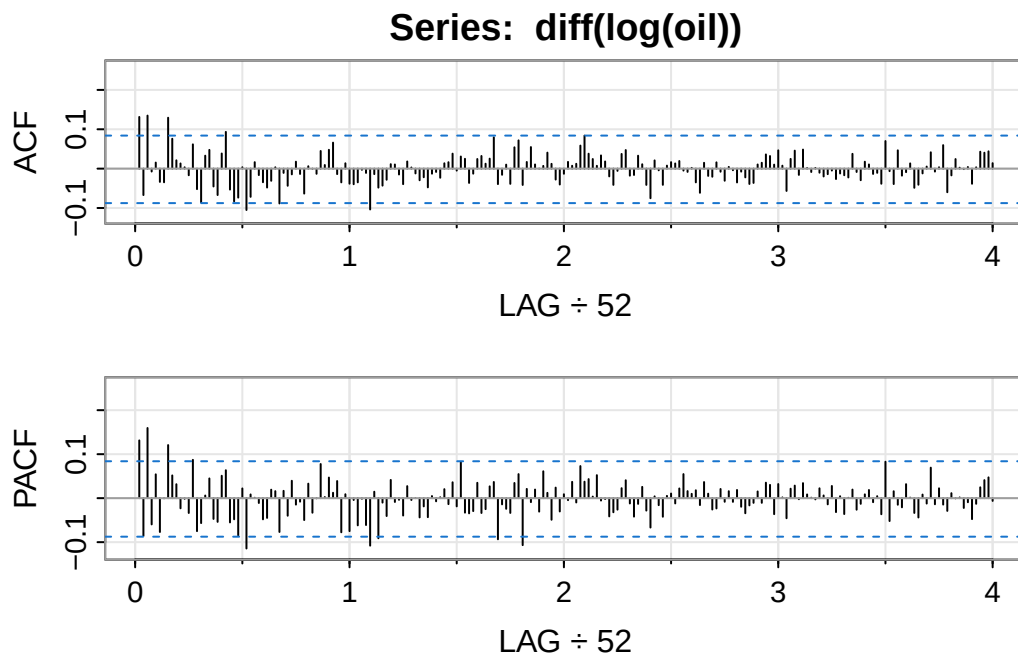


There is an upward trend, partially likely due to inflation, and a random walk component. There is also evidence of non-constant variance, so we apply a log difference like we did in 5.2.



This looks pretty close to stationarity, we note the periods of high volatility that are bunched together. This is most noticeable in 2009.

We investigate the dependence orders of the model:



	[,1]	[,2]	[,3]	[,4]	[,5]	[,6]	[,7]	[,8]	[,9]	[,10]	[,11]	[,12]	[,13]
ACF	0.13	-0.07	0.13	-0.01	0.02	-0.03	-0.03	0.13	0.08	0.02	0.01	0	-0.02
PACF	0.13	-0.09	0.16	-0.06	0.05	-0.08	0.00	0.12	0.05	0.03	-0.02	0	-0.03
	[,14]	[,15]	[,16]	[,17]	[,18]	[,19]	[,20]	[,21]	[,22]	[,23]	[,24]	[,25]	
ACF	0.06	-0.05	-0.09	0.03	0.05	-0.05	-0.07	0.04	0.09	-0.05	-0.08	-0.07	
PACF	0.09	-0.07	-0.06	0.01	0.04	-0.05	-0.05	0.05	0.06	-0.06	-0.05	-0.08	
	[,26]	[,27]	[,28]	[,29]	[,30]	[,31]	[,32]	[,33]	[,34]	[,35]	[,36]	[,37]	
ACF	0.00	-0.11	-0.07	0.02	-0.02	-0.03	-0.05	-0.03	0.00	-0.09	-0.01	-0.04	
PACF	0.02	-0.11	0.01	0.00	-0.01	-0.05	-0.04	0.02	0.02	-0.08	0.02	-0.04	
	[,38]	[,39]	[,40]	[,41]	[,42]	[,43]	[,44]	[,45]	[,46]	[,47]	[,48]	[,49]	
ACF	-0.01	0.02	-0.01	-0.06	0.01	0.00	-0.01	0.04	0.01	0.05	0.07	-0.01	
PACF	0.04	-0.01	-0.01	-0.05	0.03	-0.03	0.00	0.08	0.00	0.05	0.01	0.04	
	[,50]	[,51]	[,52]	[,53]	[,54]	[,55]	[,56]	[,57]	[,58]	[,59]	[,60]	[,61]	
ACF	-0.03	0.01	-0.04	-0.04	-0.03	0	-0.01	-0.10	-0.01	-0.05	-0.04	-0.03	
PACF	-0.08	0.01	-0.07	0.00	-0.06	0	-0.06	-0.11	0.01	-0.09	-0.01	-0.04	
	[,62]	[,63]	[,64]	[,65]	[,66]	[,67]	[,68]	[,69]	[,70]	[,71]	[,72]	[,73]	
ACF	0.01	0.01	-0.01	-0.04	0.02	0	-0.01	-0.03	-0.02	-0.05	-0.01	-0.01	
PACF	0.04	-0.01	0.00	-0.04	0.03	0	0.00	-0.04	-0.02	-0.04	0.00	-0.01	
	[,74]	[,75]	[,76]	[,77]	[,78]	[,79]	[,80]	[,81]	[,82]	[,83]	[,84]	[,85]	
ACF	-0.02	0.01	0.02	0.04	-0.01	0.03	0.02	-0.04	-0.01	0.02	0.03	0.01	
PACF	0.00	0.02	-0.01	0.04	-0.02	0.08	-0.03	-0.03	-0.03	0.03	-0.03	-0.02	
	[,86]	[,87]	[,88]	[,89]	[,90]	[,91]	[,92]	[,93]	[,94]	[,95]	[,96]	[,97]	
ACF	0.03	0.08	-0.04	-0.02	0.01	-0.04	0.05	0.07	-0.04	0.02	0.05	0.01	
PACF	0.03	0.04	-0.09	-0.01	-0.02	-0.03	0.03	0.05	-0.11	0.02	-0.01	0.02	
	[,98]	[,99]	[,100]	[,101]	[,102]	[,103]	[,104]	[,105]	[,106]	[,107]	[,108]		
ACF	0.00	0.01	0.04	0.01	-0.03	-0.04	-0.01	0.02	0.01	0.01	0.01	0.06	
PACF	-0.03	0.06	0.01	-0.05	0.02	-0.03	0.01	0.00	0.04	-0.01	0.07		
	[,109]	[,110]	[,111]	[,112]	[,113]	[,114]	[,115]	[,116]	[,117]	[,118]			
ACF	0.08	0.04	0.02	0.01	0.03	0.02	-0.02	-0.04	-0.01	0.04			
PACF	0.04	0.04	0.00	0.05	-0.01	0.00	-0.04	-0.03	-0.03	0.02			
	[,119]	[,120]	[,121]	[,122]	[,123]	[,124]	[,125]	[,126]	[,127]	[,128]			
ACF	0.05	-0.02	-0.02	0.03	0.01	-0.04	-0.08	0.02	0.00	-0.04			
PACF	0.04	-0.01	-0.04	-0.01	0.03	-0.03	-0.07	0.00	-0.02	-0.04			
	[,129]	[,130]	[,131]	[,132]	[,133]	[,134]	[,135]	[,136]	[,137]	[,138]			
ACF	0.01	0.02	0.01	0.02	0.00	-0.01	0.00	-0.03	-0.06	0.01			
PACF	0.01	0.01	-0.01	0.02	0.05	0.02	0.01	0.02	-0.02	0.04			
	[,139]	[,140]	[,141]	[,142]	[,143]	[,144]	[,145]	[,146]	[,147]	[,148]			
ACF	-0.02	-0.02	0.02	-0.01	-0.03	0.00	0.00	-0.04	-0.01	-0.02			
PACF	0.01	-0.03	-0.02	0.02	-0.01	0.02	-0.01	0.02	-0.02	-0.03			
	[,149]	[,150]	[,151]	[,152]	[,153]	[,154]	[,155]	[,156]	[,157]	[,158]			
ACF	-0.04	-0.04	0.01	0.01	0.04	0.03	0.01	0.05	0.01	-0.06			
PACF	-0.02	-0.01	0.02	-0.01	0.04	0.03	-0.04	0.03	0.00	-0.05			
	[,159]	[,160]	[,161]	[,162]	[,163]	[,164]	[,165]	[,166]	[,167]	[,168]			

ACF	0.02	0.05	-0.02	0.05	0.00	-0.01	0	-0.01	-0.02	-0.01
PACF	0.02	0.03	0.00	0.03	0.01	0.00	0	0.02	0.01	-0.01
	[,169]	[,170]	[,171]	[,172]	[,173]	[,174]	[,175]	[,176]	[,177]	[,178]
ACF	0.00	-0.03	-0.01	-0.02	-0.02	0.04	-0.01	-0.03	0.02	0.01
PACF	0.03	-0.03	0.00	-0.04	0.00	0.02	-0.03	-0.01	0.01	0.02
	[,179]	[,180]	[,181]	[,182]	[,183]	[,184]	[,185]	[,186]	[,187]	[,188]
ACF	-0.01	-0.01	-0.04	0.07	-0.01	-0.04	0.05	-0.02	-0.01	0.01
PACF	-0.01	0.00	-0.04	0.08	-0.05	0.02	-0.01	-0.02	0.03	0.00
	[,189]	[,190]	[,191]	[,192]	[,193]	[,194]	[,195]	[,196]	[,197]	[,198]
ACF	-0.05	-0.04	-0.01	0.01	0.04	-0.01	0.00	0.06	-0.06	-0.02
PACF	-0.03	-0.04	0.01	-0.01	0.07	-0.01	0.02	-0.01	-0.03	0.01
	[,199]	[,200]	[,201]	[,202]	[,203]	[,204]	[,205]	[,206]	[,207]	[,208]
ACF	0.02	0	0.00	0.00	-0.04	0.00	0.04	0.04	0.04	0.01
PACF	0.00	0	-0.02	-0.01	-0.05	-0.01	0.02	0.04	0.05	-0.01

There doesn't seem to be an obvious model to choose. Both the ACF and PACF seem to tail off to me, there aren't particularly satisfying lags to choose where they cut off. I'll start with an $ARIMA(1,1,1)$ model:

```

initial value -3.057594
iter 2 value -3.061420
iter 3 value -3.067360
iter 4 value -3.067479
iter 5 value -3.071834
iter 6 value -3.074359
iter 7 value -3.074843
iter 8 value -3.076656
iter 9 value -3.080467
iter 10 value -3.081546
iter 11 value -3.081603
iter 12 value -3.081615
iter 13 value -3.081642
iter 14 value -3.081643
iter 14 value -3.081643
iter 14 value -3.081643
final value -3.081643
converged
initial value -3.082345
iter 2 value -3.082345
iter 3 value -3.082346
iter 4 value -3.082346
iter 5 value -3.082346

```

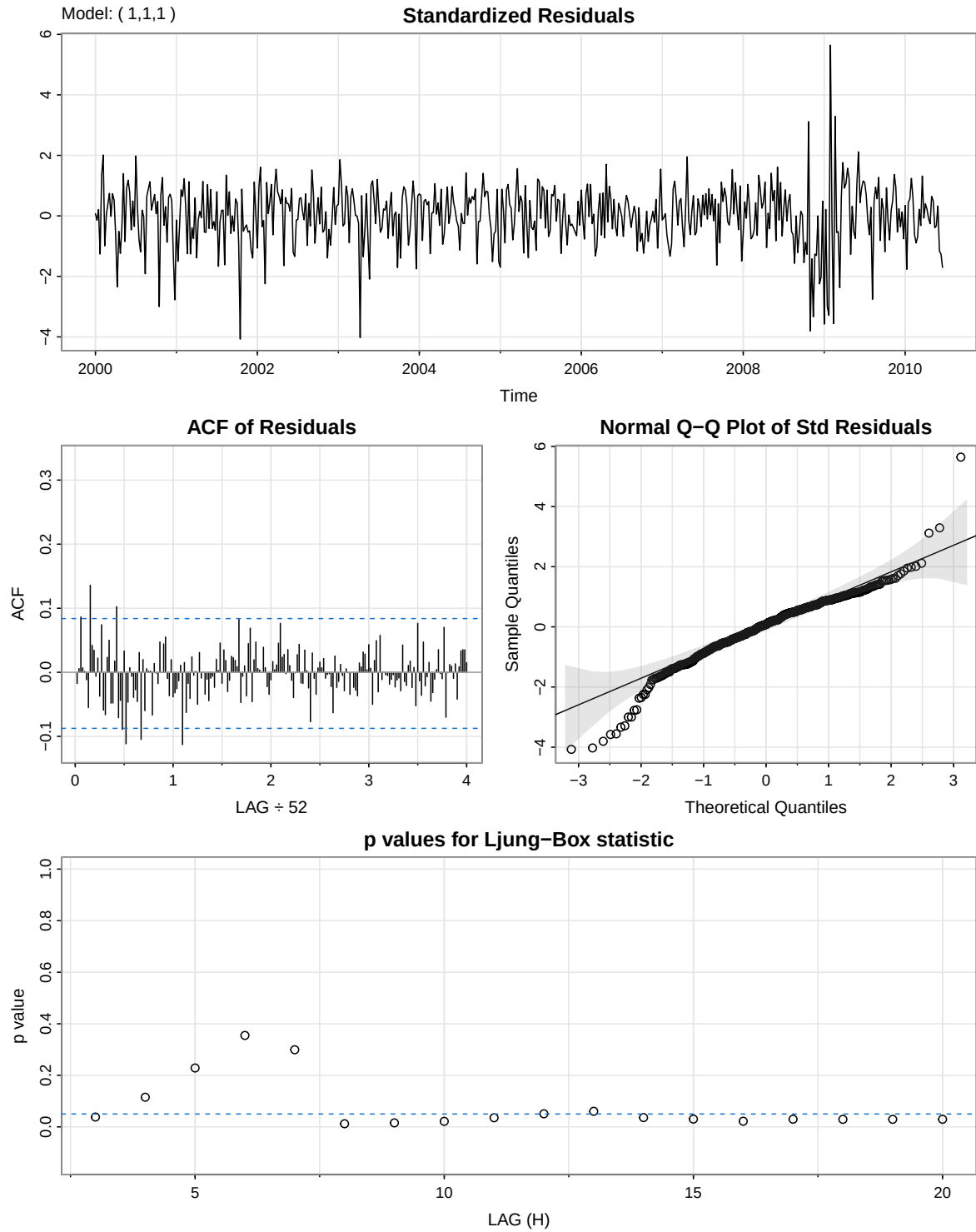
[illegible]

Coefficients:

	Estimate	SE	t.value	p.value
ar1	-0.5264	0.0871	-6.0422	0.0000
ma1	0.7146	0.0683	10.4699	0.0000
constant	0.0018	0.0022	0.7981	0.4252

sigma^2 estimated as 0.002101997 on 541 degrees of freedom

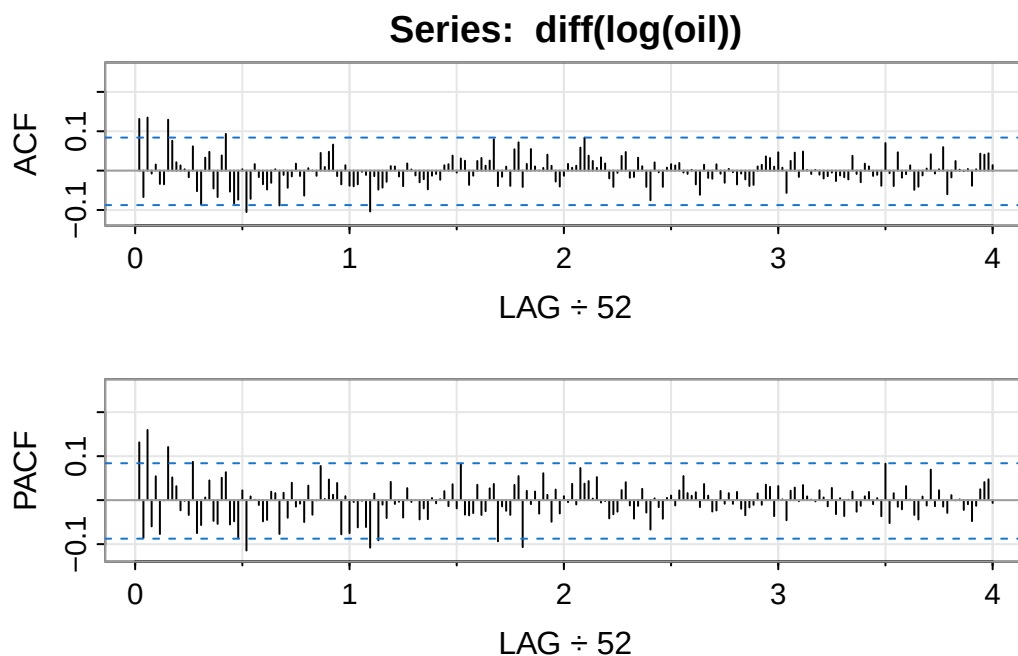
AIC = -3.312109 AICc = -3.312027 BIC = -3.280499



This fit seems inadequate, standardized residuals look stationary outside of the 2009 period.

The normal qq-plot has significant deviation in the tails. There appears to still be correlation in the residuals, as evidenced by the ACF and Ljung-Box test.

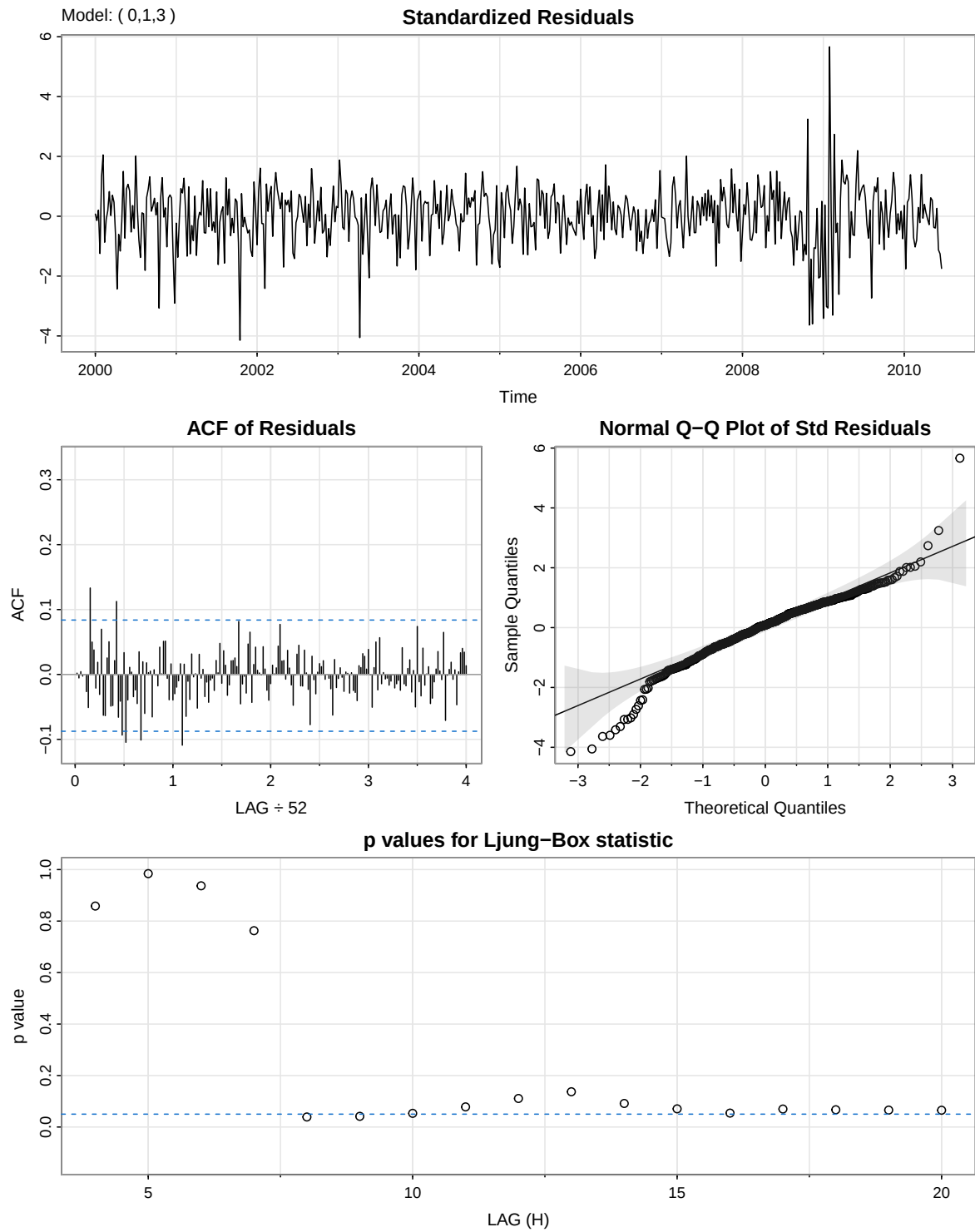
I try to fit a different model.



	[,1]	[,2]	[,3]	[,4]	[,5]	[,6]	[,7]	[,8]	[,9]	[,10]	[,11]	[,12]	[,13]
ACF	0.13	-0.07	0.13	-0.01	0.02	-0.03	-0.03	0.13	0.08	0.02	0.01	0	-0.02
PACF	0.13	-0.09	0.16	-0.06	0.05	-0.08	0.00	0.12	0.05	0.03	-0.02	0	-0.03
	[,14]	[,15]	[,16]	[,17]	[,18]	[,19]	[,20]	[,21]	[,22]	[,23]	[,24]	[,25]	
ACF	0.06	-0.05	-0.09	0.03	0.05	-0.05	-0.07	0.04	0.09	-0.05	-0.08	-0.07	
PACF	0.09	-0.07	-0.06	0.01	0.04	-0.05	-0.05	0.05	0.06	-0.06	-0.05	-0.08	
	[,26]	[,27]	[,28]	[,29]	[,30]	[,31]	[,32]	[,33]	[,34]	[,35]	[,36]	[,37]	
ACF	0.00	-0.11	-0.07	0.02	-0.02	-0.03	-0.05	-0.03	0.00	-0.09	-0.01	-0.04	
PACF	0.02	-0.11	0.01	0.00	-0.01	-0.05	-0.04	0.02	0.02	-0.08	0.02	-0.04	
	[,38]	[,39]	[,40]	[,41]	[,42]	[,43]	[,44]	[,45]	[,46]	[,47]	[,48]	[,49]	
ACF	-0.01	0.02	-0.01	-0.06	0.01	0.00	-0.01	0.04	0.01	0.05	0.07	-0.01	
PACF	0.04	-0.01	-0.01	-0.05	0.03	-0.03	0.00	0.08	0.00	0.05	0.01	0.04	
	[,50]	[,51]	[,52]	[,53]	[,54]	[,55]	[,56]	[,57]	[,58]	[,59]	[,60]	[,61]	
ACF	-0.03	0.01	-0.04	-0.04	-0.03	0	-0.01	-0.10	-0.01	-0.05	-0.04	-0.03	
PACF	-0.08	0.01	-0.07	0.00	-0.06	0	-0.06	-0.11	0.01	-0.09	-0.01	-0.04	
	[,62]	[,63]	[,64]	[,65]	[,66]	[,67]	[,68]	[,69]	[,70]	[,71]	[,72]	[,73]	
ACF	0.01	0.01	-0.01	-0.04	0.02	0	-0.01	-0.03	-0.02	-0.05	-0.01	-0.01	
PACF	0.04	-0.01	0.00	-0.04	0.03	0	0.00	-0.04	-0.02	-0.04	0.00	-0.01	

	[,74]	[,75]	[,76]	[,77]	[,78]	[,79]	[,80]	[,81]	[,82]	[,83]	[,84]	[,85]
ACF	-0.02	0.01	0.02	0.04	-0.01	0.03	0.02	-0.04	-0.01	0.02	0.03	0.01
PACF	0.00	0.02	-0.01	0.04	-0.02	0.08	-0.03	-0.03	-0.03	0.03	-0.03	-0.02
	[,86]	[,87]	[,88]	[,89]	[,90]	[,91]	[,92]	[,93]	[,94]	[,95]	[,96]	[,97]
ACF	0.03	0.08	-0.04	-0.02	0.01	-0.04	0.05	0.07	-0.04	0.02	0.05	0.01
PACF	0.03	0.04	-0.09	-0.01	-0.02	-0.03	0.03	0.05	-0.11	0.02	-0.01	0.02
	[,98]	[,99]	[,100]	[,101]	[,102]	[,103]	[,104]	[,105]	[,106]	[,107]	[,108]	
ACF	0.00	0.01	0.04	0.01	-0.03	-0.04	-0.01	0.02	0.01	0.01	0.06	
PACF	-0.03	0.06	0.01	-0.05	0.02	-0.03	0.01	0.00	0.04	-0.01	0.07	
	[,109]	[,110]	[,111]	[,112]	[,113]	[,114]	[,115]	[,116]	[,117]	[,118]		
ACF	0.08	0.04	0.02	0.01	0.03	0.02	-0.02	-0.04	-0.01	0.04		
PACF	0.04	0.04	0.00	0.05	-0.01	0.00	-0.04	-0.03	-0.03	0.02		
	[,119]	[,120]	[,121]	[,122]	[,123]	[,124]	[,125]	[,126]	[,127]	[,128]		
ACF	0.05	-0.02	-0.02	0.03	0.01	-0.04	-0.08	0.02	0.00	-0.04		
PACF	0.04	-0.01	-0.04	-0.01	0.03	-0.03	-0.07	0.00	-0.02	-0.04		
	[,129]	[,130]	[,131]	[,132]	[,133]	[,134]	[,135]	[,136]	[,137]	[,138]		
ACF	0.01	0.02	0.01	0.02	0.00	-0.01	0.00	-0.03	-0.06	0.01		
PACF	0.01	0.01	-0.01	0.02	0.05	0.02	0.01	0.02	-0.02	0.04		
	[,139]	[,140]	[,141]	[,142]	[,143]	[,144]	[,145]	[,146]	[,147]	[,148]		
ACF	-0.02	-0.02	0.02	-0.01	-0.03	0.00	0.00	-0.04	-0.01	-0.02		
PACF	0.01	-0.03	-0.02	0.02	-0.01	0.02	-0.01	0.02	-0.02	-0.03		
	[,149]	[,150]	[,151]	[,152]	[,153]	[,154]	[,155]	[,156]	[,157]	[,158]		
ACF	-0.04	-0.04	0.01	0.01	0.04	0.03	0.01	0.05	0.01	-0.06		
PACF	-0.02	-0.01	0.02	-0.01	0.04	0.03	-0.04	0.03	0.00	-0.05		
	[,159]	[,160]	[,161]	[,162]	[,163]	[,164]	[,165]	[,166]	[,167]	[,168]		
ACF	0.02	0.05	-0.02	0.05	0.00	-0.01	0	-0.01	-0.02	-0.01		
PACF	0.02	0.03	0.00	0.03	0.01	0.00	0	0.02	0.01	-0.01		
	[,169]	[,170]	[,171]	[,172]	[,173]	[,174]	[,175]	[,176]	[,177]	[,178]		
ACF	0.00	-0.03	-0.01	-0.02	-0.02	0.04	-0.01	-0.03	0.02	0.01		
PACF	0.03	-0.03	0.00	-0.04	0.00	0.02	-0.03	-0.01	0.01	0.02		
	[,179]	[,180]	[,181]	[,182]	[,183]	[,184]	[,185]	[,186]	[,187]	[,188]		
ACF	-0.01	-0.01	-0.04	0.07	-0.01	-0.04	0.05	-0.02	-0.01	0.01		
PACF	-0.01	0.00	-0.04	0.08	-0.05	0.02	-0.01	-0.02	0.03	0.00		
	[,189]	[,190]	[,191]	[,192]	[,193]	[,194]	[,195]	[,196]	[,197]	[,198]		
ACF	-0.05	-0.04	-0.01	0.01	0.04	-0.01	0.00	0.06	-0.06	-0.02		
PACF	-0.03	-0.04	0.01	-0.01	0.07	-0.01	0.02	-0.01	-0.03	0.01		
	[,199]	[,200]	[,201]	[,202]	[,203]	[,204]	[,205]	[,206]	[,207]	[,208]		
ACF	0.02	0	0.00	0.00	-0.04	0.00	0.04	0.04	0.04	0.01		
PACF	0.00	0	-0.02	-0.01	-0.05	-0.01	0.02	0.04	0.05	-0.01		

A possible other option would be a $ARIMA(0,1,3)$ model as the ACF seems to cut off after lag 3 (lag 2 doesn't appear to be significant but lag 3 is) and the PACF generally tails off.



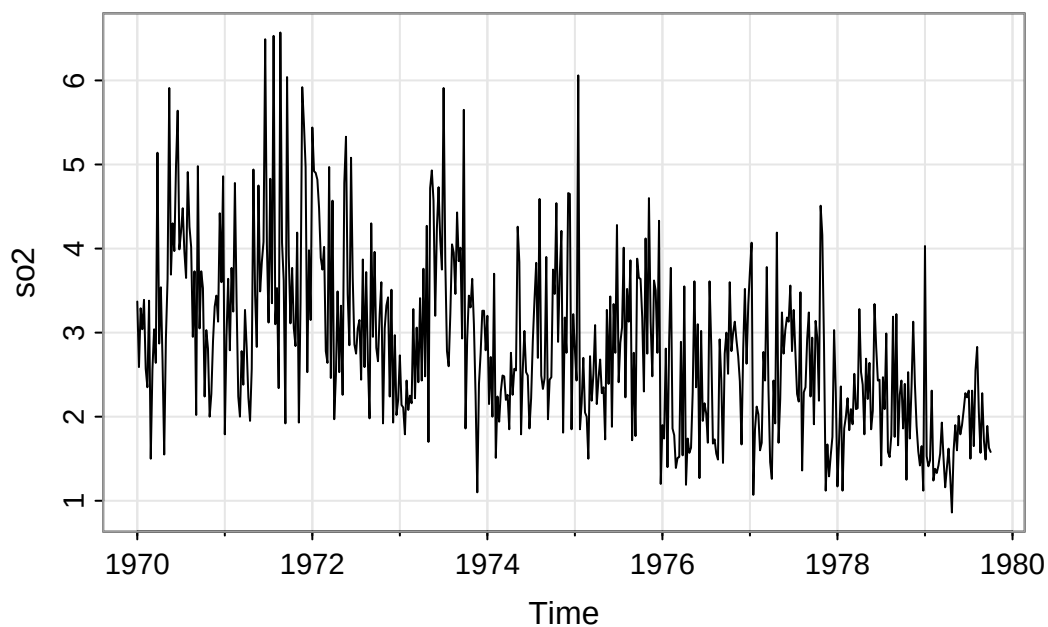
This model does a bit better than before. The standardized residual plot looks close to

stationarity. There are still deviations in normality in the tails, there might be other methods to deal with this but in the ARIMA paradigm there isn't too much we can do. Those values in 2009 seem to throw off the tails. Residuals are closer to white noise, at most early lags the Ljung-Box statistic is greater than the p-value of 0.05 and the ACF plot is close to zero. At later lags the Ljung-Box statistic sticks very close to 0.05 and the ACF seems to point toward some autocorrelation.

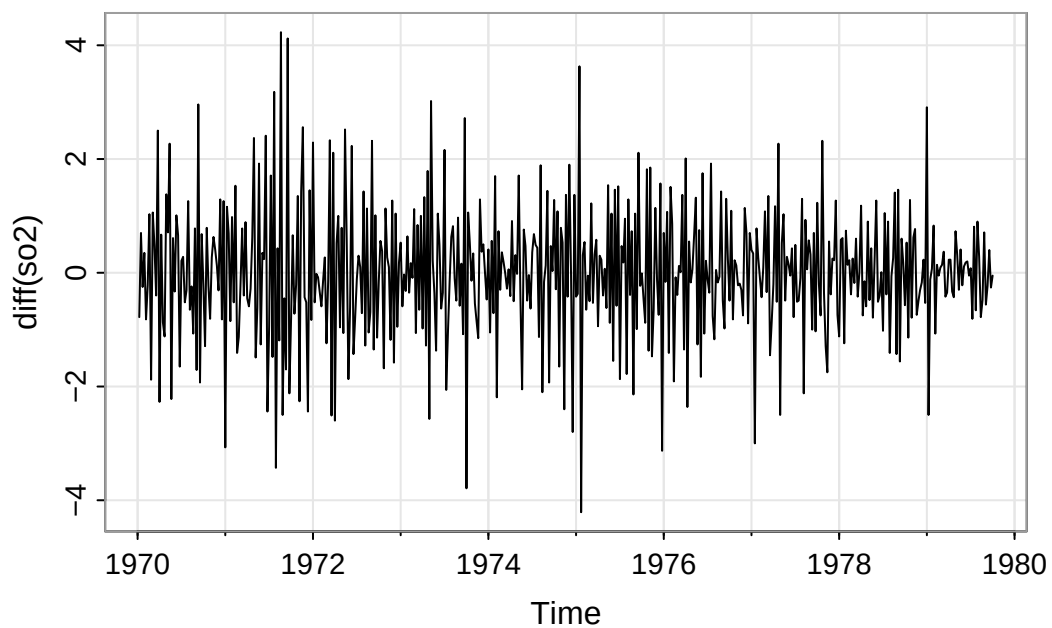
Overall the $ARIMA(0, 1, 3)$ seems to do okay, but improvements could be made if there was a model paradigm to address the non-constant variance and long tails.

5.6)

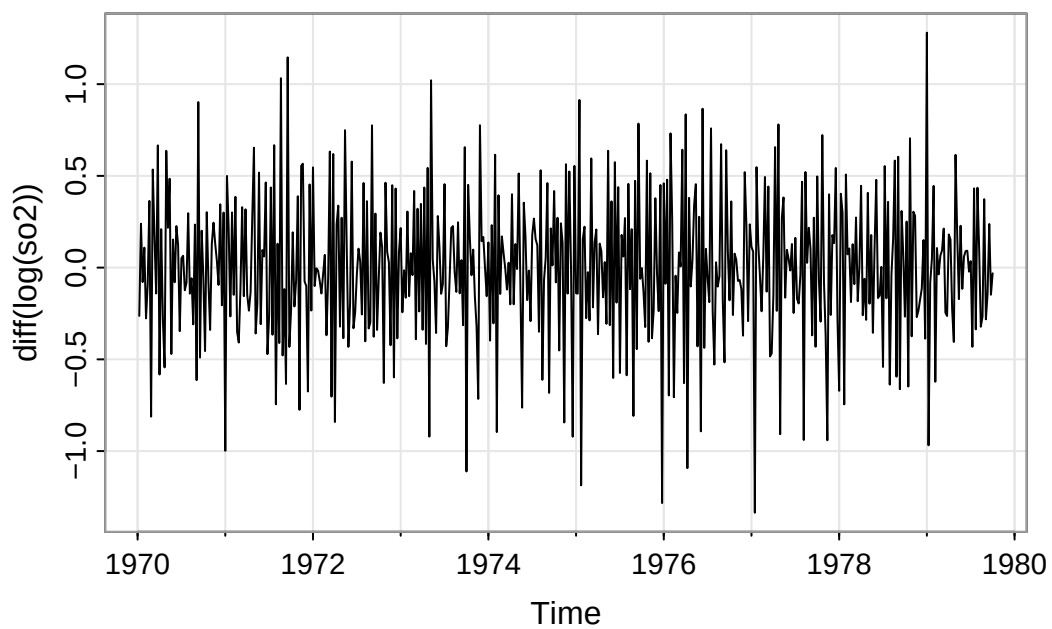
Starting with the time series plot:



There is a downward trend, so we difference.

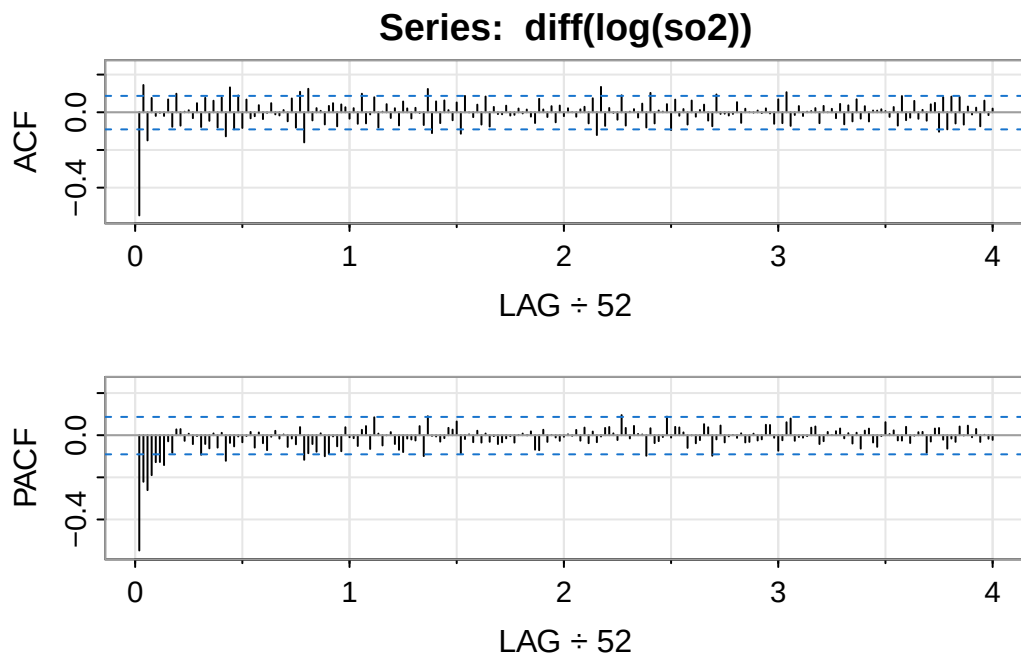


This looks close to stationarity, but there is some moderate evidence of non-constant variance. Variance seems to decrease over time. So we take a look at the plot of log-differenced SO2 data.



This looks really good, the series looks stationary. Variance now appears to be constant.

We now investigate the dependence structure:



	[,1]	[,2]	[,3]	[,4]	[,5]	[,6]	[,7]	[,8]	[,9]	[,10]	[,11]	[,12]
ACF	-0.55	0.14	-0.15	0.08	-0.02	0.00	-0.02	0.07	-0.08	0.10	-0.07	0.00
PACF	-0.55	-0.22	-0.26	-0.19	-0.13	-0.13	-0.14	-0.03	-0.08	0.03	0.03	-0.03
	[,13]	[,14]	[,15]	[,16]	[,17]	[,18]	[,19]	[,20]	[,21]	[,22]	[,23]	[,24]
ACF	0.01	-0.03	0.05	-0.08	0.08	-0.05	0.06	-0.08	0.08	-0.13	0.13	-0.09
PACF	0.01	-0.04	0.00	-0.09	-0.04	-0.06	0.01	-0.06	0.01	-0.12	-0.04	-0.05
	[,25]	[,26]	[,27]	[,28]	[,29]	[,30]	[,31]	[,32]	[,33]	[,34]	[,35]	[,36]
ACF	0.09	-0.08	0.07	-0.03	-0.02	0.04	-0.04	0.00	0.05	-0.01	-0.02	0.01
PACF	-0.01	-0.03	0.00	0.02	-0.06	0.01	-0.04	-0.07	-0.01	0.02	-0.02	0.00
	[,37]	[,38]	[,39]	[,40]	[,41]	[,42]	[,43]	[,44]	[,45]	[,46]	[,47]	[,48]
ACF	-0.05	0.07	-0.08	0.11	-0.16	0.12	-0.04	0.02	0.01	-0.06	0.03	0.05
PACF	-0.06	-0.02	-0.04	0.04	-0.12	-0.09	-0.04	-0.08	0.01	-0.10	-0.09	-0.01
	[,49]	[,50]	[,51]	[,52]	[,53]	[,54]	[,55]	[,56]	[,57]	[,58]	[,59]	[,60]
ACF	-0.07	0.04	0.03	-0.03	0.02	-0.06	0.10	-0.06	-0.01	0.08	-0.08	0.00
PACF	-0.05	-0.08	0.04	-0.01	-0.01	-0.05	0.03	0.04	-0.07	0.08	0.01	-0.05
	[,61]	[,62]	[,63]	[,64]	[,65]	[,66]	[,67]	[,68]	[,69]	[,70]	[,71]	[,72]
ACF	0.04	-0.03	0.02	-0.07	0.06	0.02	-0.03	0.02	0.00	-0.07	0.12	-0.11
PACF	0.00	0.01	-0.04	-0.07	-0.08	-0.02	-0.02	-0.02	0.04	-0.10	0.09	0.00
	[,73]	[,74]	[,75]	[,76]	[,77]	[,78]	[,79]	[,80]	[,81]	[,82]	[,83]	[,84]
ACF	0.06	-0.06	0.06	0.01	-0.04	0.05	-0.11	0.09	0	-0.03	0.04	-0.07
PACF	-0.01	-0.03	-0.01	0.03	0.03	0.07	-0.09	-0.02	0	-0.03	0.02	-0.03

	[,85]	[,86]	[,87]	[,88]	[,89]	[,90]	[,91]	[,92]	[,93]	[,94]	[,95]	[,96]
ACF	0.08	-0.08	0.03	-0.01	-0.01	0.03	-0.02	-0.01	0.02	-0.01	0.01	-0.02
PACF	0.01	-0.03	0.00	-0.04	-0.03	-0.01	0.00	-0.03	0.00	0.01	0.00	0.02
	[,97]	[,98]	[,99]	[,100]	[,101]	[,102]	[,103]	[,104]	[,105]	[,106]	[,107]	
ACF	-0.06	0.07	0.02	-0.02	0.03	-0.05	0.04	-0.02	0.02	0	-0.02	
PACF	-0.07	-0.07	0.03	-0.03	0.00	-0.01	-0.03	-0.01	0.00	0	0.03	
	[,108]	[,109]	[,110]	[,111]	[,112]	[,113]	[,114]	[,115]	[,116]	[,117]		
ACF	0.02	0.03	-0.06	0.07	-0.12	0.13	-0.07	0.02	0.00	-0.04		
PACF	-0.03	0.04	-0.04	0.02	-0.03	-0.01	0.03	0.04	0.01	-0.02		
	[,118]	[,119]	[,120]	[,121]	[,122]	[,123]	[,124]	[,125]	[,126]	[,127]		
ACF	0.09	-0.07	0.00	0.02	-0.03	0.05	-0.08	0.10	-0.06	0.01		
PACF	0.10	0.03	-0.02	0.04	0.01	0.00	-0.10	0.03	-0.04	-0.03		
	[,128]	[,129]	[,130]	[,131]	[,132]	[,133]	[,134]	[,135]	[,136]	[,137]		
ACF	0.00	0.04	-0.09	0.07	-0.02	0.02	-0.07	0.06	-0.02	0.01		
PACF	-0.01	0.09	-0.01	0.04	0.04	0.01	-0.06	0.00	-0.04	-0.02		
	[,138]	[,139]	[,140]	[,141]	[,142]	[,143]	[,144]	[,145]	[,146]	[,147]		
ACF	0.04	-0.04	-0.07	0.09	-0.01	-0.01	-0.02	-0.01	0.05	-0.06		
PACF	0.05	0.04	-0.10	-0.02	0.05	-0.03	-0.01	0.00	-0.01	0.01		
	[,148]	[,149]	[,150]	[,151]	[,152]	[,153]	[,154]	[,155]	[,156]	[,157]		
ACF	0.02	0.00	0.00	0.01	0.00	0.02	-0.01	-0.06	0.07	-0.06		
PACF	-0.03	-0.03	0.01	-0.03	-0.02	0.05	0.05	-0.01	-0.07	-0.02		
	[,158]	[,159]	[,160]	[,161]	[,162]	[,163]	[,164]	[,165]	[,166]	[,167]		
ACF	0.11	-0.07	-0.01	0.03	-0.02	0	0.01	0.02	-0.06	0.03		
PACF	0.06	0.08	-0.03	-0.01	-0.01	0	0.04	0.04	-0.04	-0.03		
	[,168]	[,169]	[,170]	[,171]	[,172]	[,173]	[,174]	[,175]	[,176]	[,177]		
ACF	0.00	0.02	-0.03	0.04	-0.06	0.04	-0.05	0.07	-0.03	0.03		
PACF	0.02	0.00	0.02	0.03	-0.01	0.01	-0.04	0.01	-0.06	0.02		
	[,178]	[,179]	[,180]	[,181]	[,182]	[,183]	[,184]	[,185]	[,186]	[,187]		
ACF	-0.05	0.01	0.01	0.02	0.01	-0.02	0.03	-0.07	0.09	-0.04		
PACF	0.03	-0.03	-0.06	0.00	0.06	0.01	0.02	-0.02	-0.03	0.04		
	[,188]	[,189]	[,190]	[,191]	[,192]	[,193]	[,194]	[,195]	[,196]	[,197]		
ACF	-0.03	0.06	-0.03	0.01	-0.04	0.04	0.05	-0.10	0.08	-0.09		
PACF	-0.04	0.00	0.01	0.02	-0.08	-0.03	0.03	0.03	-0.02	-0.06		
	[,198]	[,199]	[,200]	[,201]	[,202]	[,203]	[,204]	[,205]	[,206]	[,207]		
ACF	0.08	-0.06	0.08	-0.07	0.03	-0.01	0.02	-0.07	0.06	-0.01		
PACF	-0.02	-0.03	0.04	0.00	0.05	-0.01	0.03	-0.03	0.00	-0.02		
	[,208]											
ACF	0.02											
PACF	-0.02											

It seems both ACF and PACF tail off, so we start with an $ARIMA(1,1,1)$ model. There could also be an argument for $ARIMA(0,1,1)$ because the second and third lags are pretty

close to being insignificant. Or an $ARIMA(0,1,3)$ model because ACF drops close to zero after the third lag.

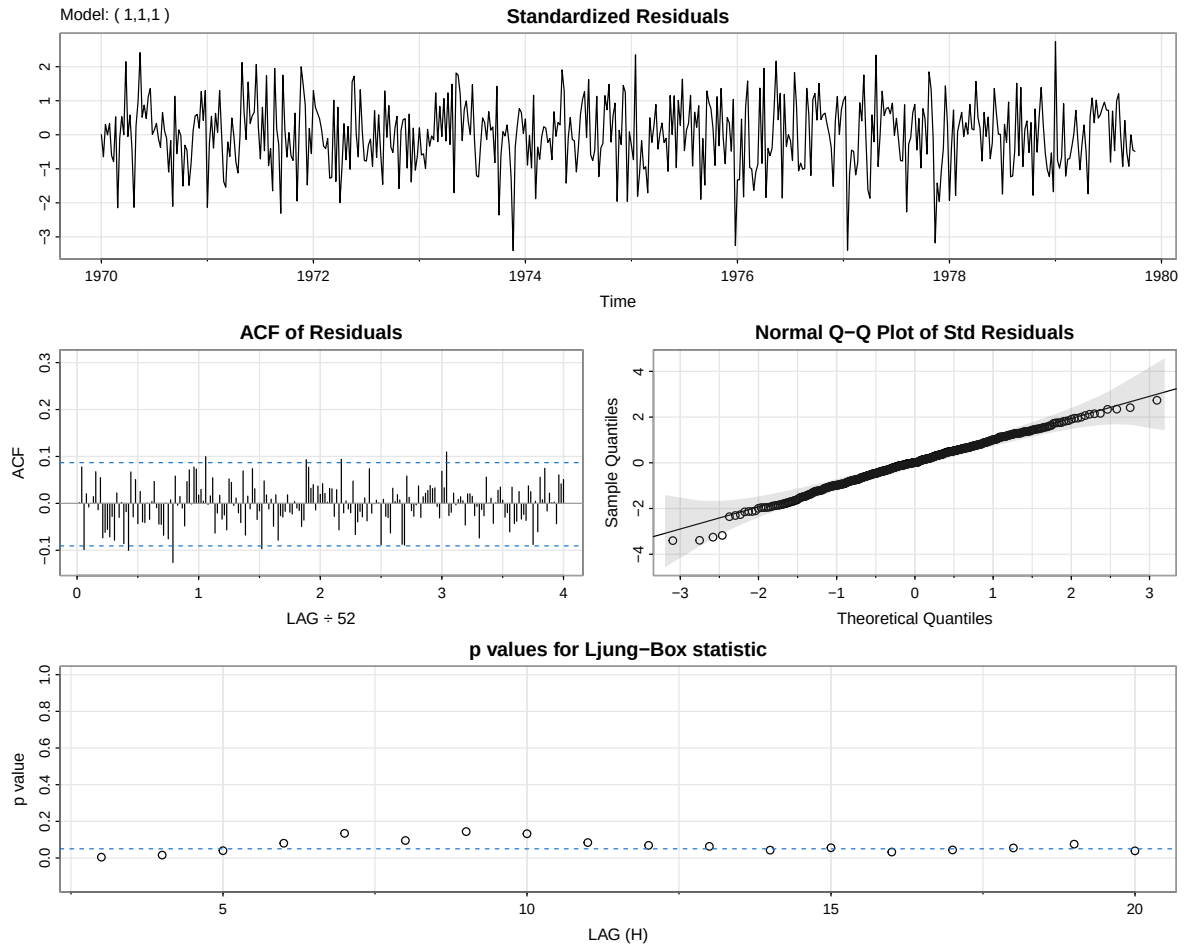
```
initial    value -0.900031  
iter      2 value -1.099284  
iter      3 value -1.142829  
iter      4 value -1.165771  
iter      5 value -1.173248  
iter      6 value -1.175085  
iter      7 value -1.175638  
iter      8 value -1.175807  
iter      9 value -1.175813  
iter     10 value -1.175815  
iter     11 value -1.175817  
iter     12 value -1.175818  
iter     12 value -1.175818  
iter     12 value -1.175818  
final     value -1.175818  
converged  
  
initial    value -1.175719  
iter      2 value -1.175722  
iter      3 value -1.175724  
iter      4 value -1.175726  
iter      5 value -1.175726  
iter      6 value -1.175726  
iter      6 value -1.175726  
iter      6 value -1.175726  
iter      6 value -1.175726  
final     value -1.175726  
converged
```

Coefficients:

	Estimate	SE	t.value	p.value
ar1	-0.0338	0.0576	-0.5866	0.5577
ma1	-0.8233	0.0381	-21.5993	0.0000
constant	-0.0010	0.0024	-0.4252	0.6708

sigma^2 estimated as 0.09500773 on 504 degrees of freedom

AIC = 0.5022039 AICc = 0.502298 BIC = 0.5355649

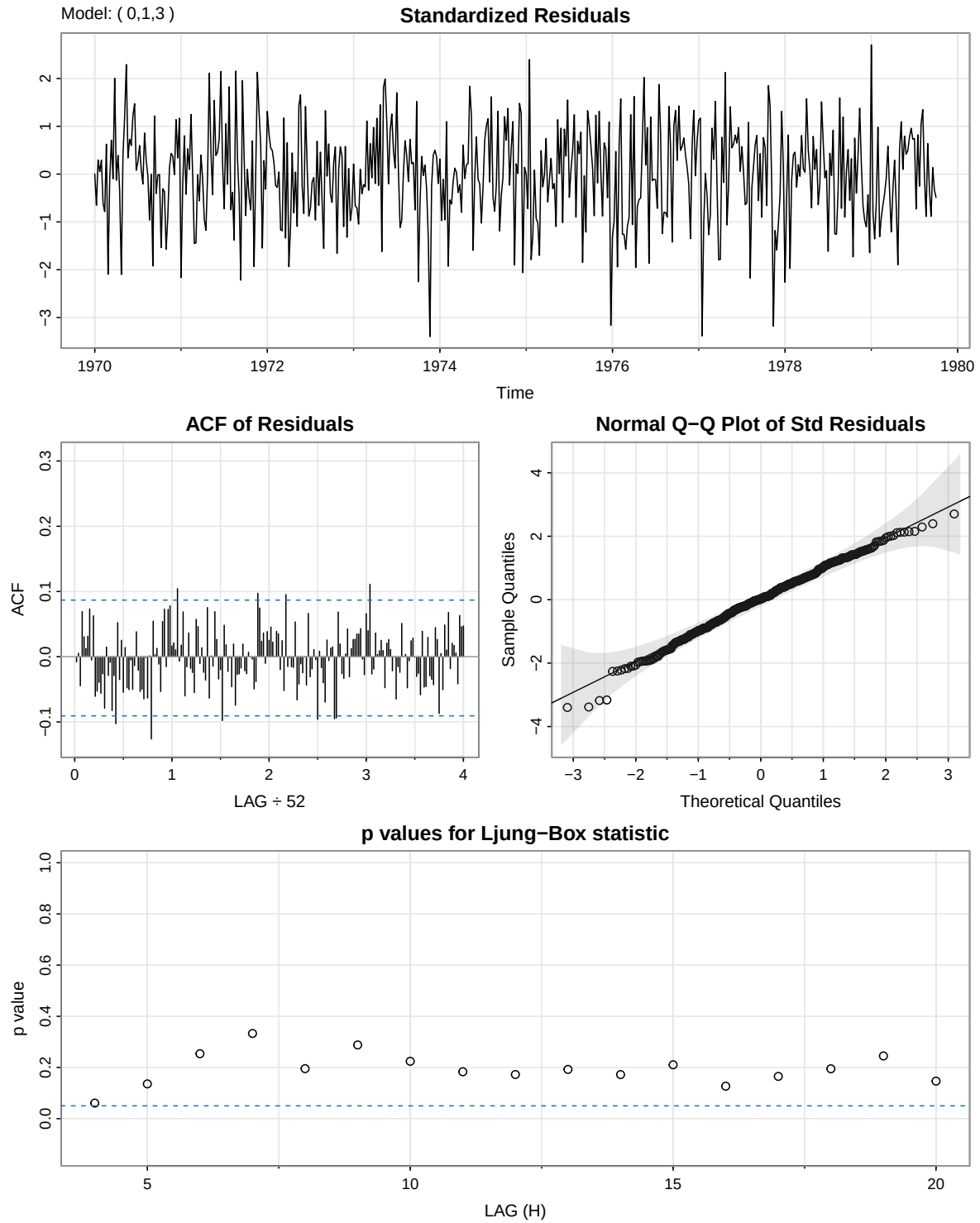


The standardized residuals look stationary, close to normally distributed with constant variance. There is evidence of autocorrelation remaining at shorter lags, and at yearly intervals. In this problem we are tasked to only fit an ARIMA model, so we try $ARIMA(0, 1, 3)$:

```

initial value -0.900609
iter 2 value -1.117182
iter 3 value -1.170182
iter 4 value -1.177812
iter 5 value -1.179007
iter 6 value -1.181490
iter 7 value -1.182237
iter 8 value -1.182265
iter 9 value -1.182268
iter 10 value -1.182268
iter 10 value -1.182268

```

This model keeps the good aspects of before, and improves significantly on the autocorrelation

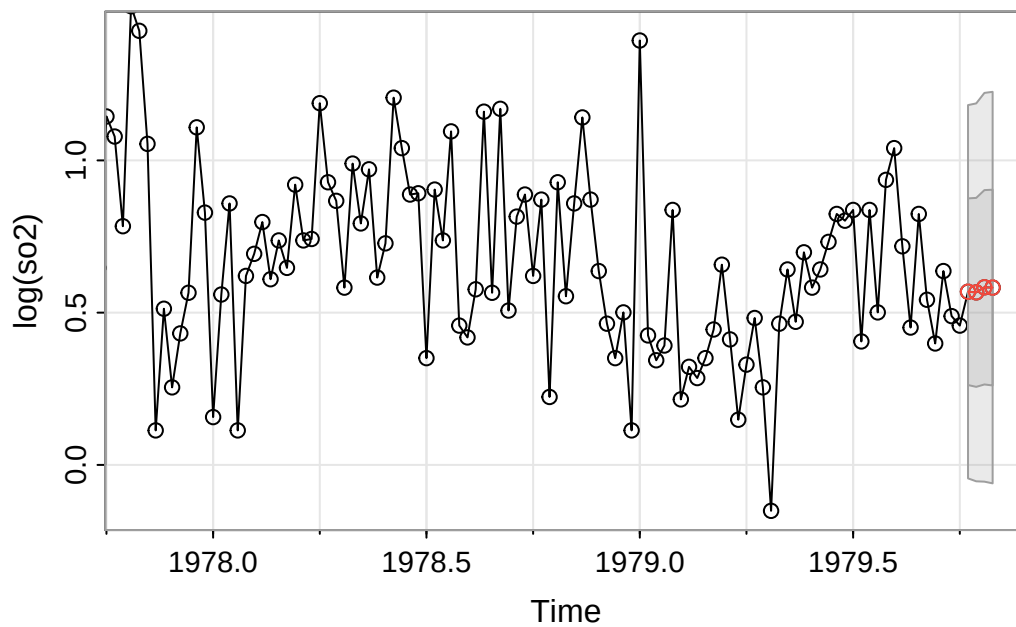
in residuals. They now appear to be white noise.

A comparison of AIC/BIC between the two models is:

Model	AIC	BIC
$ARIMA(1, 1, 1)$	.502	.536
$ARIMA(0, 1, 3)$	.495	.537

AIC would choose $ARIMA(0, 1, 3)$, BIC would choose $ARIMA(1, 1, 1)$. I would select $ARIMA(0, 1, 3)$ because diagnostics are improved and AIC prefers it.

A forecast of four time periods is plotted below:



```
$pred
Time Series:
Start = c(1979, 41)
End = c(1979, 44)
Frequency = 52
[1] 0.5687078 0.5666028 0.5833113 0.5822776
```

```
$se
Time Series:
Start = c(1979, 41)
```

```

End = c(1979, 44)
Frequency = 52
[1] 0.3064603 0.3102616 0.3191452 0.3214473

```

The 95% confidence intervals are:

Time Series:

```
Start = c(1979, 41)
```

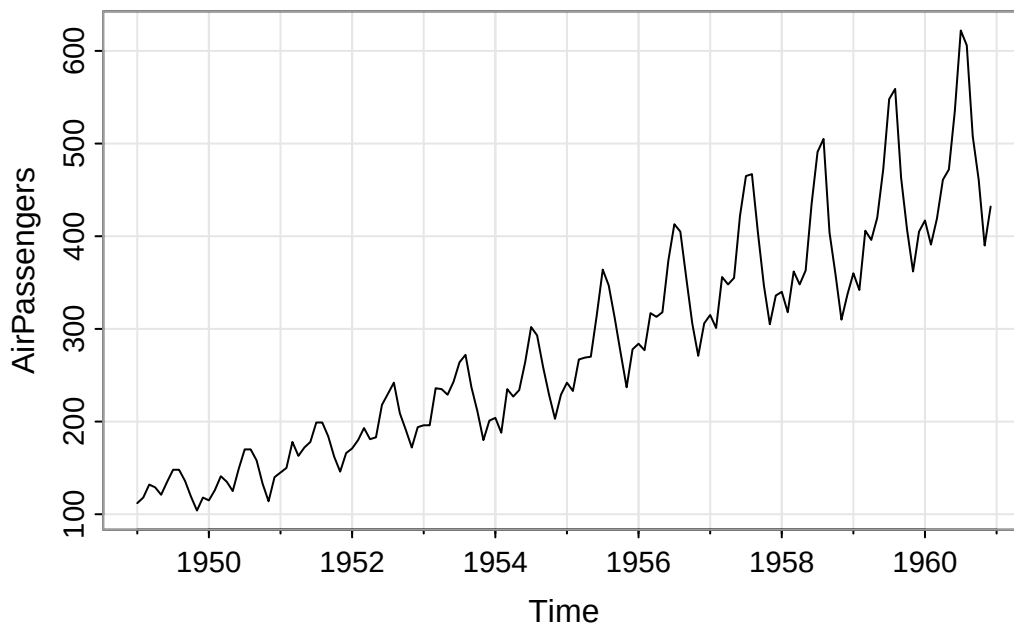
```
End = c(1979, 44)
```

```
Frequency = 52
```

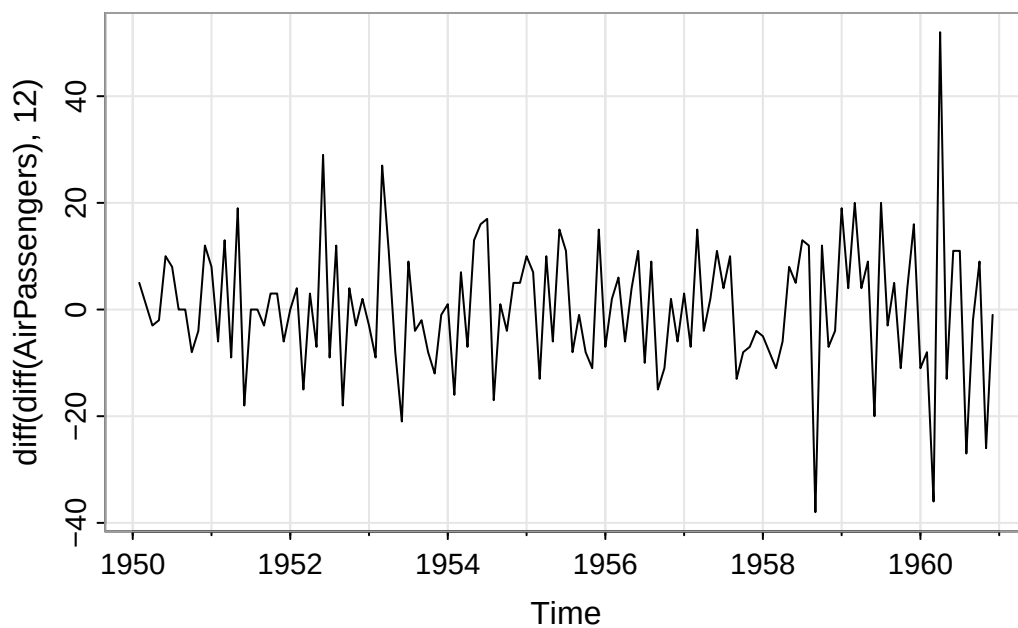
	point - 1.96 * se	point	point + 1.96 * se
1979.769	-0.03195441	0.5687078	1.169370
1979.788	-0.04150999	0.5666028	1.174716
1979.808	-0.04221333	0.5833113	1.208836
1979.827	-0.04775916	0.5822776	1.212314

5.7)

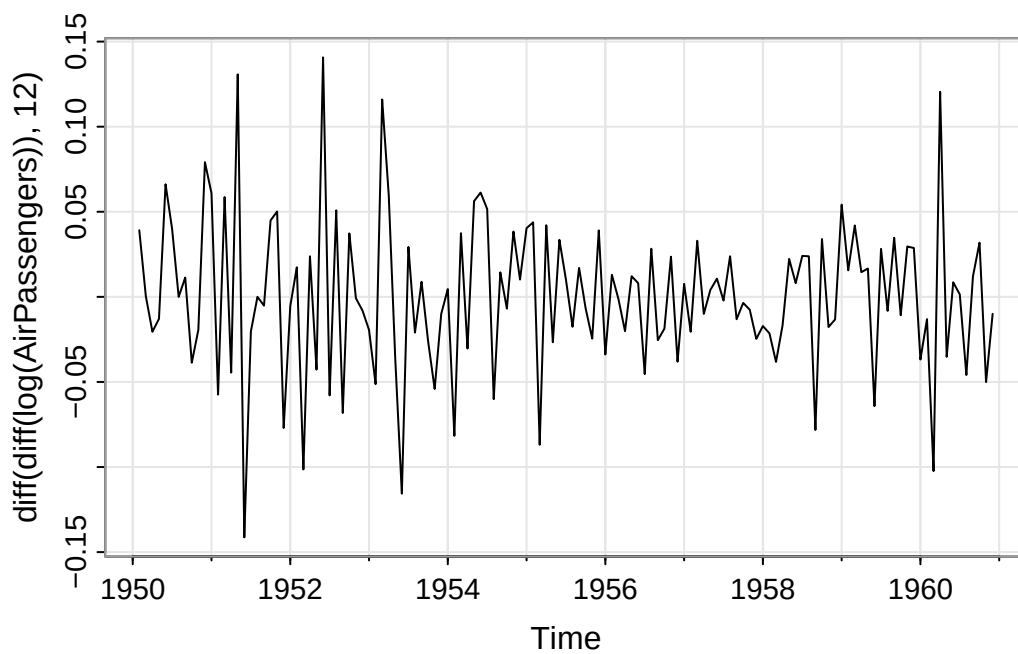
We start with a time series plot:



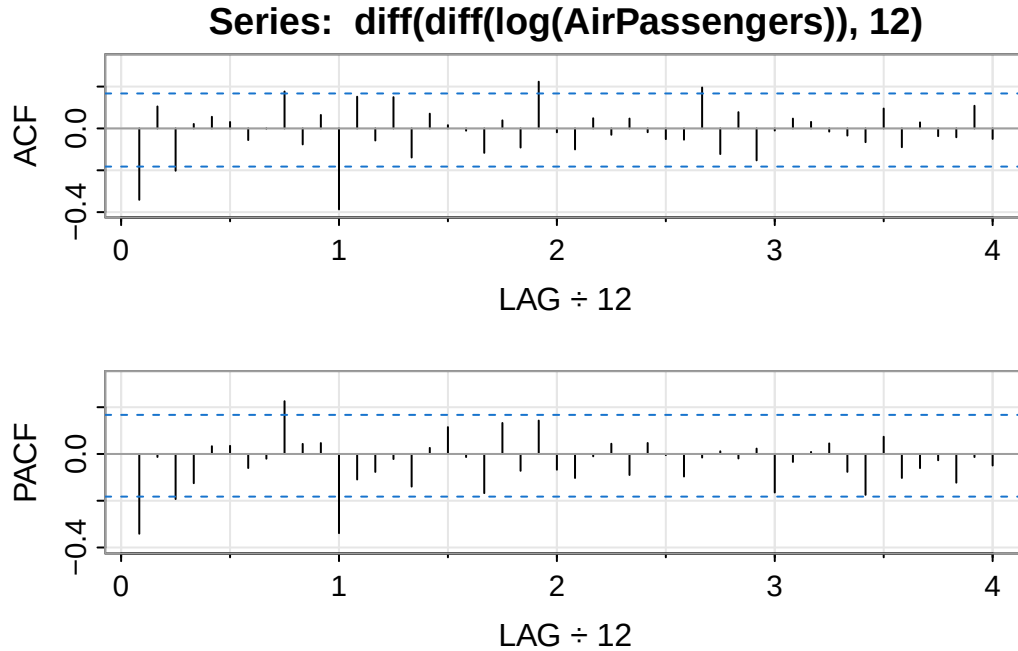
There is an upward trend, and what appears to be yearly seasonality. We first order difference and difference over 12 months.



There appears to be some evidence of non-constant variance, variance increases over time. In particular, after 1958 and twice in 1960 we see large moves that are not replicated in earlier portions of the series. So we apply a log transformation, and then difference again.



This looks closer to constant variance, but there are several large changes early in the series. We stick with this as it looks stationary, and investigate the dependence structure.



	[,1]	[,2]	[,3]	[,4]	[,5]	[,6]	[,7]	[,8]	[,9]	[,10]	[,11]	[,12]	[,13]
ACF	-0.34	0.11	-0.20	0.02	0.06	0.03	-0.06	0.00	0.18	-0.08	0.06	-0.39	0.15
	[,14]	[,15]	[,16]	[,17]	[,18]	[,19]	[,20]	[,21]	[,22]	[,23]	[,24]	[,25]	
PACF	-0.34	-0.01	-0.19	-0.13	0.03	0.03	-0.06	-0.02	0.23	0.04	0.05	-0.34	-0.11
ACF	-0.06	0.15	-0.14	0.07	0.02	-0.01	-0.12	0.04	-0.09	0.22	-0.02	-0.1	
PACF	-0.08	-0.02	-0.14	0.03	0.11	-0.01	-0.17	0.13	-0.07	0.14	-0.07	-0.1	
	[,26]	[,27]	[,28]	[,29]	[,30]	[,31]	[,32]	[,33]	[,34]	[,35]	[,36]	[,37]	
ACF	0.05	-0.03	0.05	-0.02	-0.05	-0.05	0.20	-0.12	0.08	-0.15	-0.01	0.05	
PACF	-0.01	0.04	-0.09	0.05	0.00	-0.10	-0.02	0.01	-0.02	0.02	-0.16	-0.03	
	[,38]	[,39]	[,40]	[,41]	[,42]	[,43]	[,44]	[,45]	[,46]	[,47]	[,48]		
ACF	0.03	-0.02	-0.03	-0.07	0.10	-0.09	0.03	-0.04	-0.04	0.11	-0.05		
PACF	0.01	0.05	-0.08	-0.17	0.07	-0.10	-0.06	-0.03	-0.12	-0.01	-0.05		

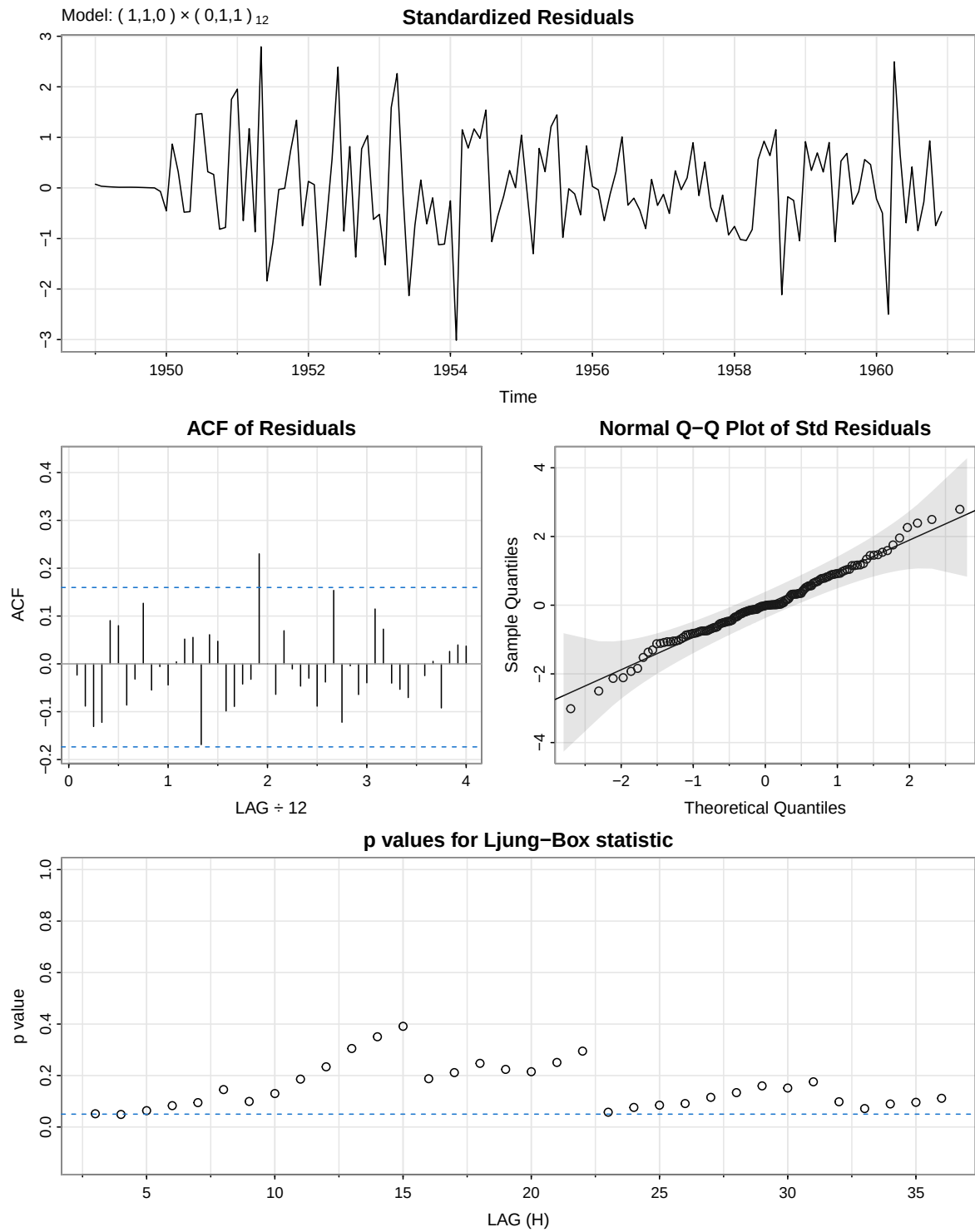
First we look at the within seasonal effect by focusing on early lags. The ACF and PACF have similar behavior, both drop below significance after the first lag but become significant again at the third lag. So its not obvious how to move forward. One possible choice would be $p = 1$ and $q = 0$ since the PACF drops close to zero after the first lag and the ACF could be argued to tail off.

So our first model is $ARIMA(1, 1, 0, 0, 1, 1)_{12}$ on the log-transformed SO2 data.

Coefficients:

σ^2 estimated as 0.00136738 on 129 degrees of freedom

29



Standardized residuals have close to constant variance and are normally distributed. The ACF

plot looks good at early lags, but some later lags jump out. The Ljung-Box statistic suggests at lower lags there is still some autocorrelation.

While this model is close, its worth considering a competitor.

The seasonal behavior seems to be well contained, so I don't change that. Rather I select $p = 1$, $q = 1$ because it can be argued that both the ACF and PACF tail off outside of auto-correlations close to 0 at lag 2.

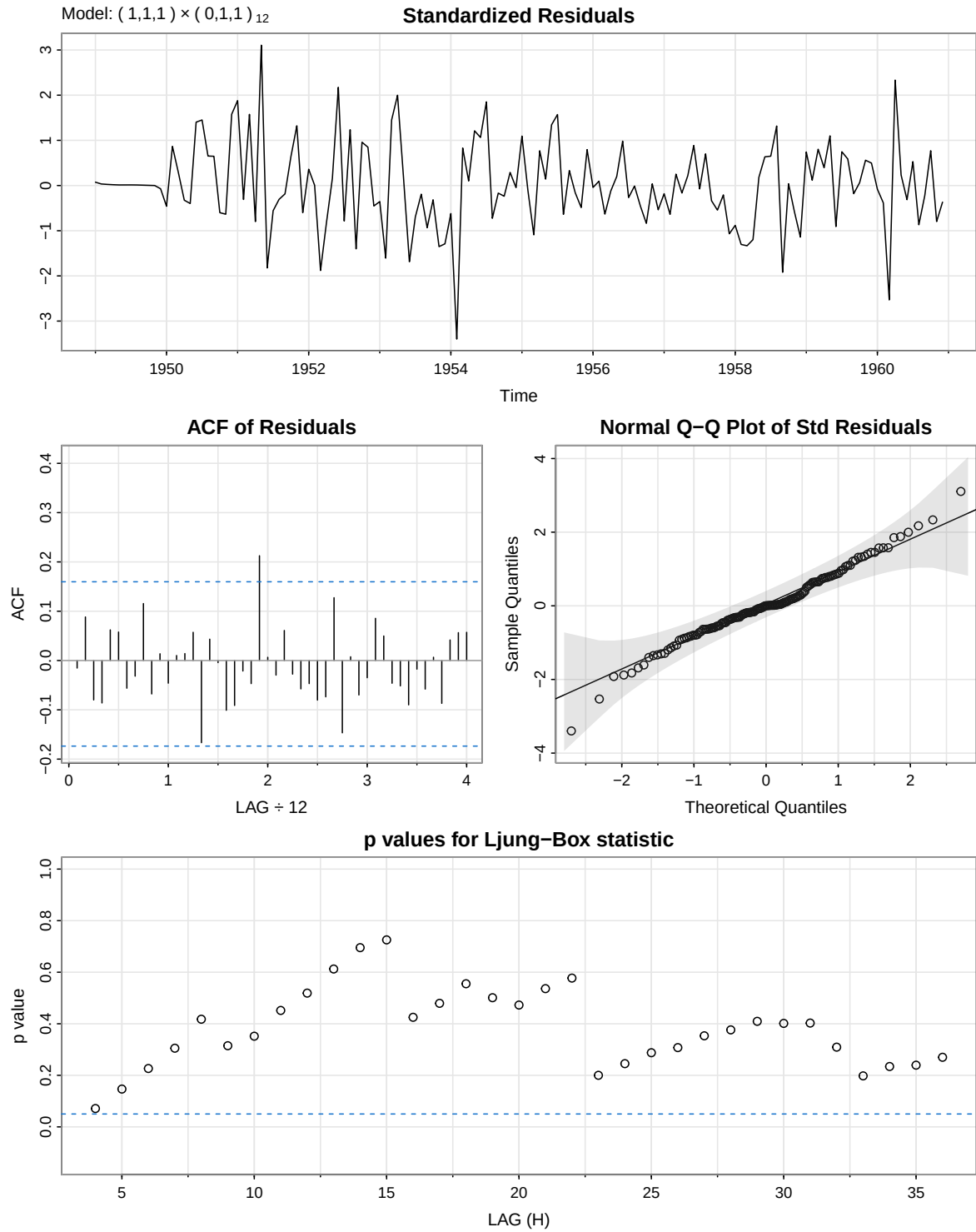
[illegible]

Coefficients:

	Estimate	SE	t.value	p.value
ar1	0.1960	0.2475	0.7921	0.4298
ma1	-0.5784	0.2132	-2.7127	0.0076
sma1	-0.5643	0.0747	-7.5544	0.0000

σ^2 estimated as 0.001341097 on 128 degrees of freedom

AIC = -3.678622 AICc = -3.677179 BIC = -3.59083



Residuals are similar, but we see improvement in auto-correlation at early lags in the Ljung-

Box plot. I use AIC/BIC to help decide.

Model	AIC	BIC
$ARIMA(1, 1, 0, 0, 1, 1)_{12}$	-3.675	-3.610
$ARIMA(1, 1, 1, 0, 1, 1)_{12}$	-3.679	-3.591

The AIC and BIC disagree, AIC chooses $ARIMA(1, 1, 1, 0, 1, 1)_{12}$ and BIC chooses $ARIMA(1, 1, 0, 0, 1, 1)_{12}$. I go with $ARIMA(1, 1, 0, 0, 1, 1)_{12}$ because the ACF of residuals at early lags doesn't seem too concerning, and generally we prefer simpler models.

5.17)

The $ARIMA(0, 1, 1)$ model can be written as:

$$x_t = x_{t-1} + w_t - \lambda w_{t-1}$$

If we have $y_t = x_t - x_{t-1}$ we have:

$$y_t = x_{t-1} + w_t - \lambda w_{t-1} - x_{t-1} = w_t - \lambda w_{t-1}$$

So, in terms of y_i this is a MA(1) model. We can represent the invertible form:

$$w_t = \sum_{j=0}^{\infty} \lambda^j y_{t-j}$$

If we sub in $y_t = x_t - x_{t-1}$:

$$\begin{aligned} w_t &= \sum_{j=0}^{\infty} \lambda^j (x_{t-j} - x_{t-j-1}) = \sum_{j=0}^{\infty} \lambda^j x_{t-j} - \sum_{j=0}^{\infty} \lambda^j x_{t-j-1} = x_t + \sum_{j=1}^{\infty} \lambda^j x_{t-j} - \sum_{j=0}^{\infty} \lambda^j x_{t-j-1} \\ &= x_t + \sum_{j=1}^{\infty} \lambda^j x_{t-j} - \sum_{j=1}^{\infty} \lambda^j x_{t-j-1} + x_{t-1} \end{aligned}$$

Got a little stuck at this point, but once you get to 5.24:

$$x_t = \sum_{j=1}^{\infty} (1 - \lambda) \lambda^{j-1} x_{t-j} + w_t$$

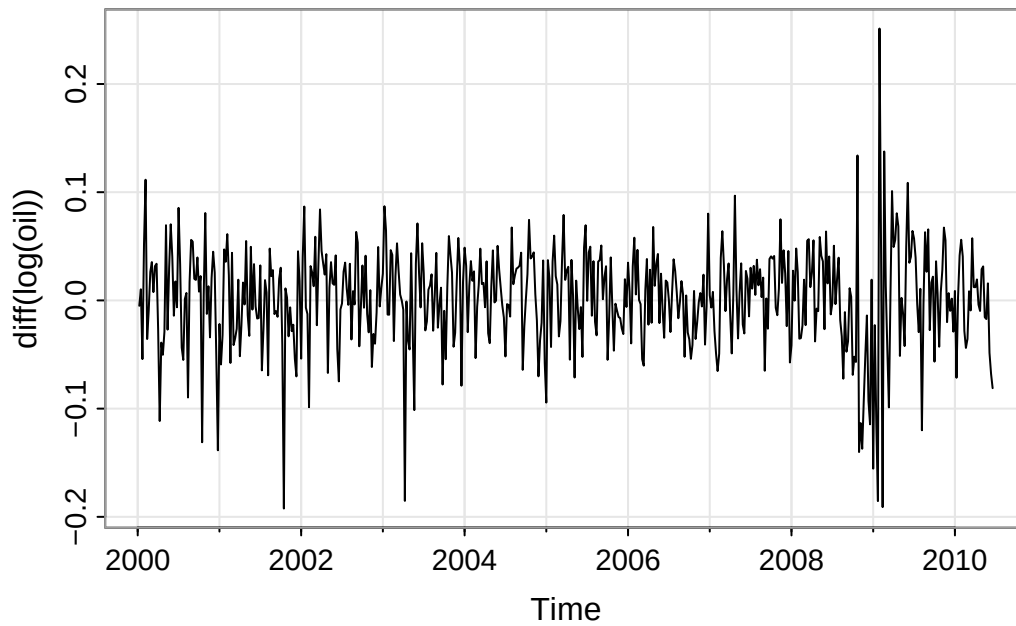
We can show:

$$x_{n+1}^n = \sum_{j=1}^{\infty} (1-\lambda)\lambda^{j-1}x_{n+1-j}^n = (1-\lambda)x_n^n + \lambda x_n^{n-1}$$

By using infinite geometric sum facts and backsubstitution. The w_t term drops because we are taking expected value.

8.3)

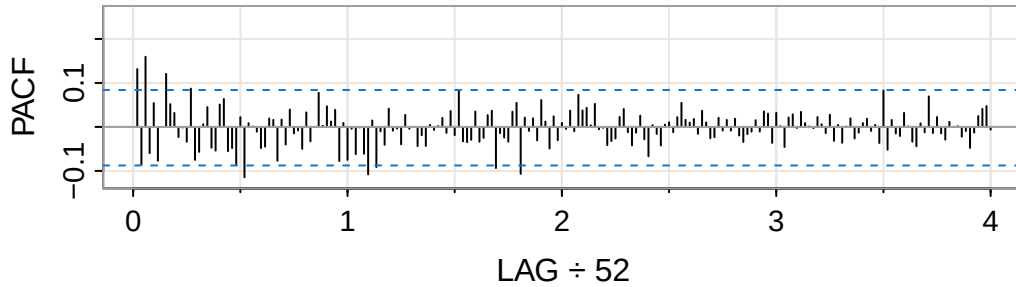
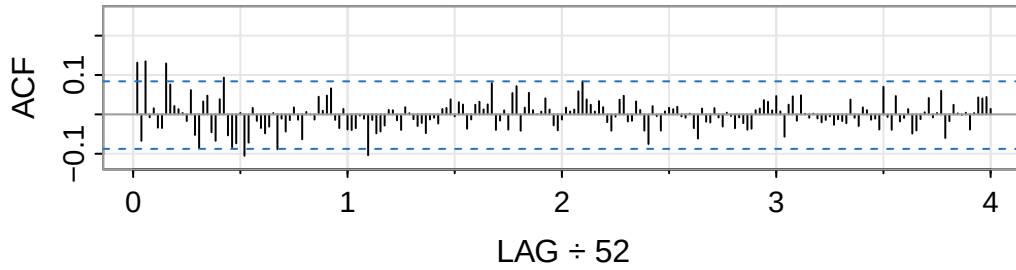
We start with a time series plot of the log differenced (returns) of oil.



The tell tale sign of GARCH behavior is clusters of volatility, this is clearly present in the data and is most easily seen at the start of 2009.

To start we look at the P/ACF of the returns:

Series: diff(log(oil))



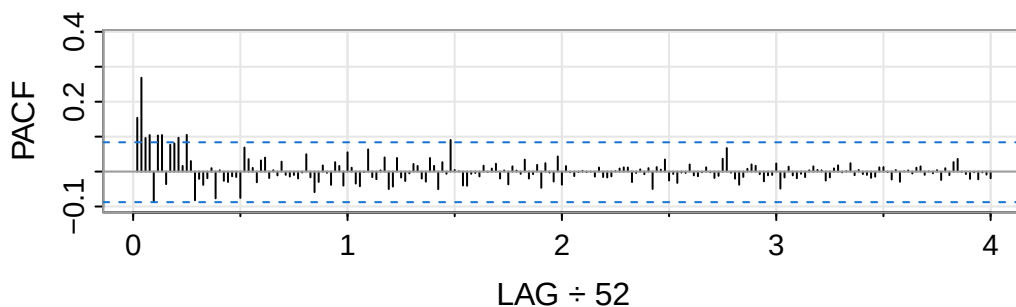
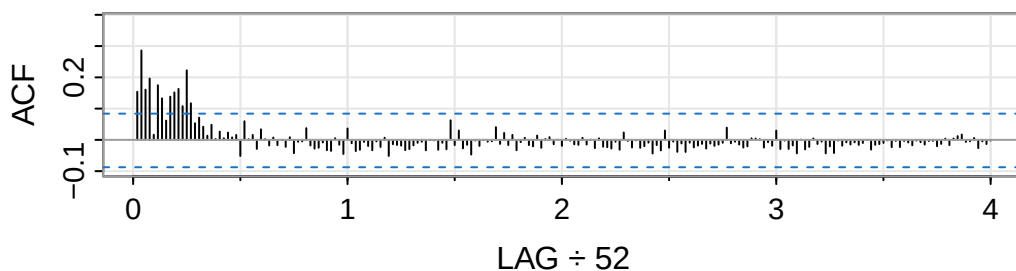
	[,1]	[,2]	[,3]	[,4]	[,5]	[,6]	[,7]	[,8]	[,9]	[,10]	[,11]	[,12]	[,13]
ACF	0.13	-0.07	0.13	-0.01	0.02	-0.03	-0.03	0.13	0.08	0.02	0.01	0	-0.02
PACF	0.13	-0.09	0.16	-0.06	0.05	-0.08	0.00	0.12	0.05	0.03	-0.02	0	-0.03
	[,14]	[,15]	[,16]	[,17]	[,18]	[,19]	[,20]	[,21]	[,22]	[,23]	[,24]	[,25]	
ACF	0.06	-0.05	-0.09	0.03	0.05	-0.05	-0.07	0.04	0.09	-0.05	-0.08	-0.07	
PACF	0.09	-0.07	-0.06	0.01	0.04	-0.05	-0.05	0.05	0.06	-0.06	-0.05	-0.08	
	[,26]	[,27]	[,28]	[,29]	[,30]	[,31]	[,32]	[,33]	[,34]	[,35]	[,36]	[,37]	
ACF	0.00	-0.11	-0.07	0.02	-0.02	-0.03	-0.05	-0.03	0.00	-0.09	-0.01	-0.04	
PACF	0.02	-0.11	0.01	0.00	-0.01	-0.05	-0.04	0.02	0.02	-0.08	0.02	-0.04	
	[,38]	[,39]	[,40]	[,41]	[,42]	[,43]	[,44]	[,45]	[,46]	[,47]	[,48]	[,49]	
ACF	-0.01	0.02	-0.01	-0.06	0.01	0.00	-0.01	0.04	0.01	0.05	0.07	-0.01	
PACF	0.04	-0.01	-0.01	-0.05	0.03	-0.03	0.00	0.08	0.00	0.05	0.01	0.04	
	[,50]	[,51]	[,52]	[,53]	[,54]	[,55]	[,56]	[,57]	[,58]	[,59]	[,60]	[,61]	
ACF	-0.03	0.01	-0.04	-0.04	-0.03	0	-0.01	-0.10	-0.01	-0.05	-0.04	-0.03	
PACF	-0.08	0.01	-0.07	0.00	-0.06	0	-0.06	-0.11	0.01	-0.09	-0.01	-0.04	
	[,62]	[,63]	[,64]	[,65]	[,66]	[,67]	[,68]	[,69]	[,70]	[,71]	[,72]	[,73]	
ACF	0.01	0.01	-0.01	-0.04	0.02	0	-0.01	-0.03	-0.02	-0.05	-0.01	-0.01	
PACF	0.04	-0.01	0.00	-0.04	0.03	0	0.00	-0.04	-0.02	-0.04	0.00	-0.01	
	[,74]	[,75]	[,76]	[,77]	[,78]	[,79]	[,80]	[,81]	[,82]	[,83]	[,84]	[,85]	
ACF	-0.02	0.01	0.02	0.04	-0.01	0.03	0.02	-0.04	-0.01	0.02	0.03	0.01	
PACF	0.00	0.02	-0.01	0.04	-0.02	0.08	-0.03	-0.03	-0.03	0.03	-0.03	-0.02	
	[,86]	[,87]	[,88]	[,89]	[,90]	[,91]	[,92]	[,93]	[,94]	[,95]	[,96]	[,97]	
ACF	0.03	0.08	-0.04	-0.02	0.01	-0.04	0.05	0.07	-0.04	0.02	0.05	0.01	

PACF	0.03	0.04	-0.09	-0.01	-0.02	-0.03	0.03	0.05	-0.11	0.02	-0.01	0.02
	[,98]	[,99]	[,100]	[,101]	[,102]	[,103]	[,104]	[,105]	[,106]	[,107]	[,108]	
ACF	0.00	0.01	0.04	0.01	-0.03	-0.04	-0.01	0.02	0.01	0.01	0.01	0.06
PACF	-0.03	0.06	0.01	-0.05	0.02	-0.03	0.01	0.00	0.04	-0.01	0.07	
	[,109]	[,110]	[,111]	[,112]	[,113]	[,114]	[,115]	[,116]	[,117]	[,118]		
ACF	0.08	0.04	0.02	0.01	0.03	0.02	-0.02	-0.04	-0.01	0.04		
PACF	0.04	0.04	0.00	0.05	-0.01	0.00	-0.04	-0.03	-0.03	0.02		
	[,119]	[,120]	[,121]	[,122]	[,123]	[,124]	[,125]	[,126]	[,127]	[,128]		
ACF	0.05	-0.02	-0.02	0.03	0.01	-0.04	-0.08	0.02	0.00	-0.04		
PACF	0.04	-0.01	-0.04	-0.01	0.03	-0.03	-0.07	0.00	-0.02	-0.04		
	[,129]	[,130]	[,131]	[,132]	[,133]	[,134]	[,135]	[,136]	[,137]	[,138]		
ACF	0.01	0.02	0.01	0.02	0.00	-0.01	0.00	-0.03	-0.06	0.01		
PACF	0.01	0.01	-0.01	0.02	0.05	0.02	0.01	0.02	-0.02	0.04		
	[,139]	[,140]	[,141]	[,142]	[,143]	[,144]	[,145]	[,146]	[,147]	[,148]		
ACF	-0.02	-0.02	0.02	-0.01	-0.03	0.00	0.00	-0.04	-0.01	-0.02		
PACF	0.01	-0.03	-0.02	0.02	-0.01	0.02	-0.01	0.02	-0.02	-0.03		
	[,149]	[,150]	[,151]	[,152]	[,153]	[,154]	[,155]	[,156]	[,157]	[,158]		
ACF	-0.04	-0.04	0.01	0.01	0.04	0.03	0.01	0.05	0.01	-0.06		
PACF	-0.02	-0.01	0.02	-0.01	0.04	0.03	-0.04	0.03	0.00	-0.05		
	[,159]	[,160]	[,161]	[,162]	[,163]	[,164]	[,165]	[,166]	[,167]	[,168]		
ACF	0.02	0.05	-0.02	0.05	0.00	-0.01	0	-0.01	-0.02	-0.01		
PACF	0.02	0.03	0.00	0.03	0.01	0.00	0	0.02	0.01	-0.01		
	[,169]	[,170]	[,171]	[,172]	[,173]	[,174]	[,175]	[,176]	[,177]	[,178]		
ACF	0.00	-0.03	-0.01	-0.02	-0.02	0.04	-0.01	-0.03	0.02	0.01		
PACF	0.03	-0.03	0.00	-0.04	0.00	0.02	-0.03	-0.01	0.01	0.02		
	[,179]	[,180]	[,181]	[,182]	[,183]	[,184]	[,185]	[,186]	[,187]	[,188]		
ACF	-0.01	-0.01	-0.04	0.07	-0.01	-0.04	0.05	-0.02	-0.01	0.01		
PACF	-0.01	0.00	-0.04	0.08	-0.05	0.02	-0.01	-0.02	0.03	0.00		
	[,189]	[,190]	[,191]	[,192]	[,193]	[,194]	[,195]	[,196]	[,197]	[,198]		
ACF	-0.05	-0.04	-0.01	0.01	0.04	-0.01	0.00	0.06	-0.06	-0.02		
PACF	-0.03	-0.04	0.01	-0.01	0.07	-0.01	0.02	-0.01	-0.03	0.01		
	[,199]	[,200]	[,201]	[,202]	[,203]	[,204]	[,205]	[,206]	[,207]	[,208]		
ACF	0.02	0	0.00	0.00	-0.04	0.00	0.04	0.04	0.04	0.01		
PACF	0.00	0	-0.02	-0.01	-0.05	-0.01	0.02	0.04	0.05	-0.01		

We previously chose $p = 0, q = 3$ when we got our best fit of the model possible. We stick with that for now.

Next we look at the P/ACF of the centered squared returns $(r_t - \hat{\mu}_t)^2$. Following the example of 8.1, we plot the squared residuals from our $MA(3)$ model.

Series: ma_3_resids^2



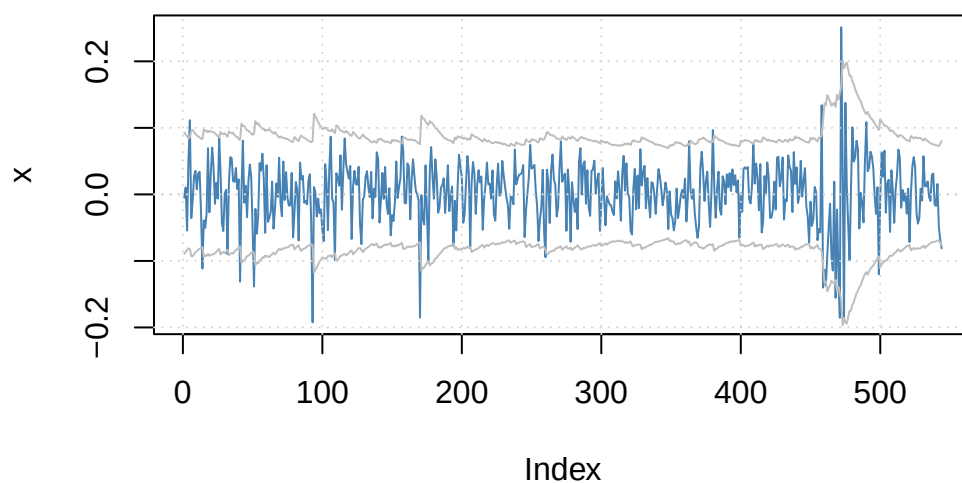
	[,1]	[,2]	[,3]	[,4]	[,5]	[,6]	[,7]	[,8]	[,9]	[,10]	[,11]	[,12]	[,13]
ACF	0.15	0.29	0.16	0.2	0.02	0.17	0.13	0.06	0.14	0.15	0.16	0.11	0.22
PACF	0.15	0.27	0.10	0.1	-0.09	0.10	0.10	-0.04	0.08	0.08	0.10	0.02	0.10
	[,14]	[,15]	[,16]	[,17]	[,18]	[,19]	[,20]	[,21]	[,22]	[,23]	[,24]	[,25]	
ACF	0.12	0.05	0.07	0.04	0.01	0.05	0.00	0.03	0.01	0.02	0.01	0.02	
PACF	0.03	-0.08	-0.02	-0.04	-0.02	0.01	-0.08	0.00	-0.03	-0.03	-0.01	-0.02	
	[,26]	[,27]	[,28]	[,29]	[,30]	[,31]	[,32]	[,33]	[,34]	[,35]	[,36]	[,37]	
ACF	-0.05	0.06	0.00	0.02	-0.03	0.03	0.00	-0.02	0.01	-0.02	0.00	-0.02	
PACF	-0.08	0.07	0.04	0.01	-0.03	0.03	0.04	-0.02	0.00	-0.01	0.03	-0.01	
	[,38]	[,39]	[,40]	[,41]	[,42]	[,43]	[,44]	[,45]	[,46]	[,47]	[,48]	[,49]	
ACF	0.01	-0.04	-0.01	0	0.04	-0.02	-0.03	-0.03	-0.01	-0.03	-0.04	0.01	
PACF	-0.01	-0.01	-0.02	0	0.05	-0.02	-0.06	-0.03	0.02	0.00	-0.04	0.03	
	[,50]	[,51]	[,52]	[,53]	[,54]	[,55]	[,56]	[,57]	[,58]	[,59]	[,60]	[,61]	
ACF	-0.02	-0.05	0.04	-0.01	-0.04	-0.03	-0.01	-0.02	-0.03	0.00	-0.02	0.01	
PACF	0.02	-0.04	0.06	0.01	-0.03	-0.04	0.00	0.06	-0.02	-0.02	0.00	0.04	
	[,62]	[,63]	[,64]	[,65]	[,66]	[,67]	[,68]	[,69]	[,70]	[,71]	[,72]	[,73]	
ACF	-0.05	-0.01	-0.02	-0.02	-0.03	-0.03	-0.02	-0.01	-0.01	-0.03	0.00	0.00	
PACF	-0.05	-0.04	0.04	-0.02	-0.03	-0.01	0.02	0.02	-0.02	-0.03	0.04	0.02	
	[,74]	[,75]	[,76]	[,77]	[,78]	[,79]	[,80]	[,81]	[,82]	[,83]	[,84]	[,85]	
ACF	-0.03	-0.01	-0.03	0.06	-0.02	0.03	-0.03	-0.02	-0.05	0	-0.02	0.00	
PACF	-0.05	0.03	-0.01	0.09	0.00	0.00	-0.04	-0.04	-0.01	0	-0.01	0.02	
	[,86]	[,87]	[,88]	[,89]	[,90]	[,91]	[,92]	[,93]	[,94]	[,95]	[,96]	[,97]	
ACF	-0.01	0.00	0.04	-0.01	0.02	-0.02	0.02	-0.03	-0.01	0.01	-0.02	-0.02	

PACF	0.00	0.01	0.02	-0.02	0.00	-0.04	0.01	0.00	-0.01	0.03	-0.02	-0.01
	[,98]	[,99]	[,100]	[,101]	[,102]	[,103]	[,104]	[,105]	[,106]	[,107]	[,108]	
ACF	0.01	-0.03	0.00	0.01	-0.01	0.00	-0.02	0.00	0	-0.01	-0.02	
PACF	0.02	-0.05	0.02	0.00	-0.03	0.04	-0.04	0.02	0	-0.01	0.00	
	[,109]	[,110]	[,111]	[,112]	[,113]	[,114]	[,115]	[,116]	[,117]	[,118]		
ACF	0.01	-0.01	0	-0.03	0.01	-0.02	-0.02	-0.03	-0.01	-0.03		
PACF	0.00	0.00	0	-0.01	0.01	-0.02	-0.02	-0.01	0.00	0.01		
	[,119]	[,120]	[,121]	[,122]	[,123]	[,124]	[,125]	[,126]	[,127]	[,128]		
ACF	0.02	0.00	-0.02	0	-0.03	-0.02	-0.01	-0.04	-0.02	-0.03		
PACF	0.01	0.01	-0.03	0	0.01	-0.01	0.01	-0.05	0.01	0.01		
	[,129]	[,130]	[,131]	[,132]	[,133]	[,134]	[,135]	[,136]	[,137]	[,138]		
ACF	0.03	-0.02	-0.01	-0.04	-0.01	-0.04	-0.01	-0.02	-0.02	-0.01		
PACF	0.03	-0.03	-0.01	-0.03	0.00	0.00	0.02	-0.01	-0.01	0.00		
	[,139]	[,140]	[,141]	[,142]	[,143]	[,144]	[,145]	[,146]	[,147]	[,148]		
ACF	-0.03	-0.01	-0.02	-0.02	-0.01	0.04	-0.01	-0.01	-0.01	-0.02		
PACF	0.01	0.00	-0.02	-0.01	0.04	0.07	0.00	-0.02	-0.04	-0.02		
	[,149]	[,150]	[,151]	[,152]	[,153]	[,154]	[,155]	[,156]	[,157]	[,158]		
ACF	-0.02	0.00	0.00	0.00	-0.02	-0.01	-0.02	0.03	-0.03	0.00		
PACF	0.01	0.02	0.02	-0.01	-0.03	-0.01	-0.01	0.02	-0.05	-0.02		
	[,159]	[,160]	[,161]	[,162]	[,163]	[,164]	[,165]	[,166]	[,167]	[,168]		
ACF	-0.03	-0.02	-0.04	0.00	-0.03	-0.02	0.00	-0.01	-0.01	-0.04		
PACF	0.01	-0.01	-0.02	-0.01	-0.02	0.00	0.01	0.00	0.00	-0.03		
	[,169]	[,170]	[,171]	[,172]	[,173]	[,174]	[,175]	[,176]	[,177]	[,178]		
ACF	-0.02	-0.04	0.00	-0.02	-0.01	-0.01	-0.01	-0.02	-0.01	0.00		
PACF	-0.02	0.01	0.02	0.00	0.00	0.02	-0.01	0.00	-0.01	-0.01		
	[,179]	[,180]	[,181]	[,182]	[,183]	[,184]	[,185]	[,186]	[,187]	[,188]		
ACF	-0.03	-0.02	-0.01	-0.01	0	-0.02	0	-0.02	0	-0.01		
PACF	-0.02	-0.01	0.01	0.01	0	-0.02	0	-0.03	0	0.00		
	[,189]	[,190]	[,191]	[,192]	[,193]	[,194]	[,195]	[,196]	[,197]	[,198]		
ACF	-0.02	0.00	-0.01	-0.01	0	0	-0.02	-0.01	0.00	-0.02		
PACF	-0.01	0.01	0.01	-0.01	0	0	0.00	-0.02	0.01	-0.01		
	[,199]	[,200]	[,201]	[,202]	[,203]	[,204]	[,205]	[,206]	[,207]	[,208]		
ACF	0.00	0.01	0.02	-0.01	0.00	0.01	-0.03	0	-0.01	0.00		
PACF	0.03	0.04	0.00	-0.01	-0.02	0.00	-0.02	0	-0.01	-0.02		

Both the ACF and PACF of the squared residuals are large, confirming the GARCH effect, and tail off suggesting that we should use a GARCH(1,1) extension. Additionally, we have seen that errors were not normal in our previous attempt so we choose t-errors.

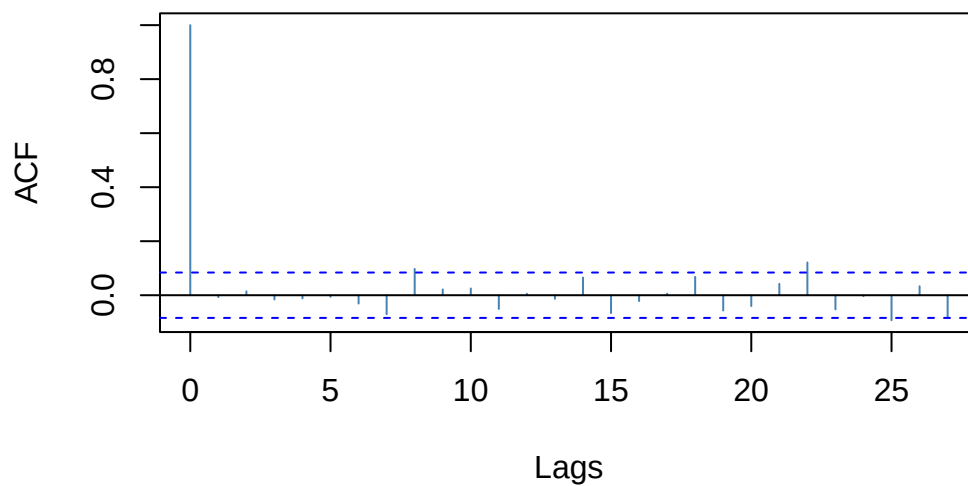
We can see the impact of the conditional SD over that end of 2008-2009 period:

Series with 2 Conditional SD Superimposed

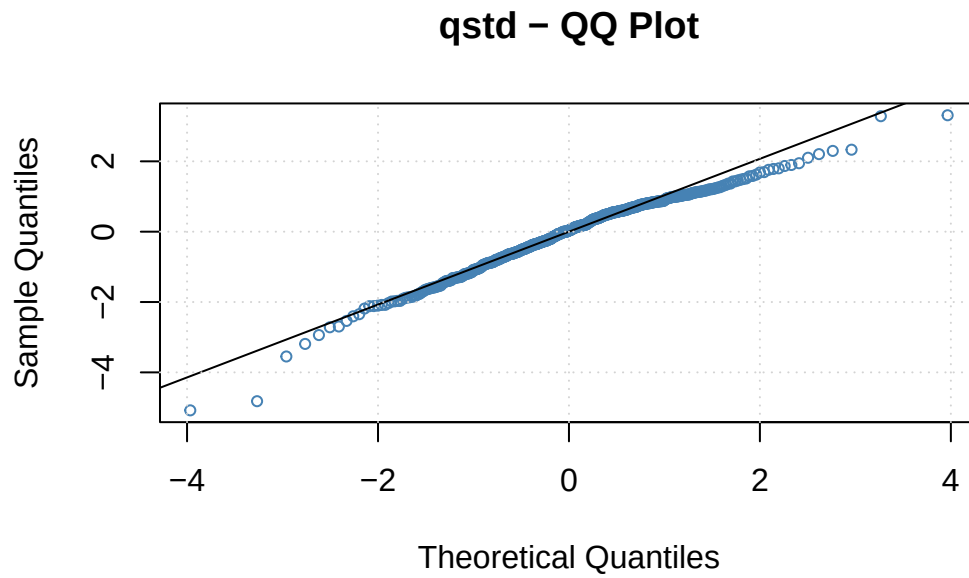


The model responds to the increased volatility effectively.

ACF of Standardized Residuals



The ACF of residuals shows that we now have really good white noise behavior.

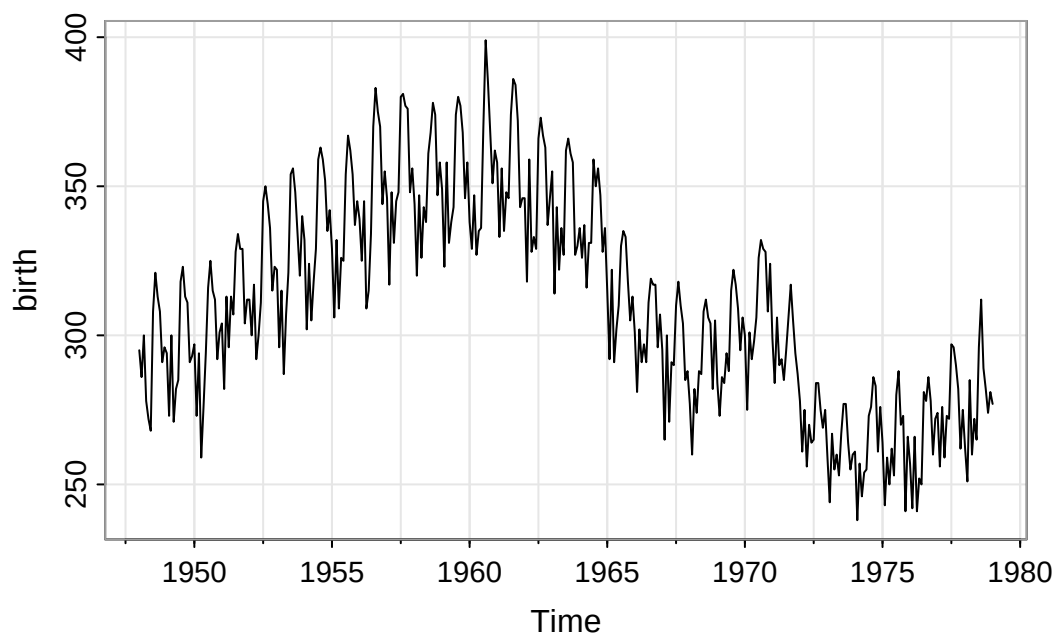


The standardized t distribution QQ-plot shows that residuals fit that distribution quite well with only a few deviations in the tail areas.

II.

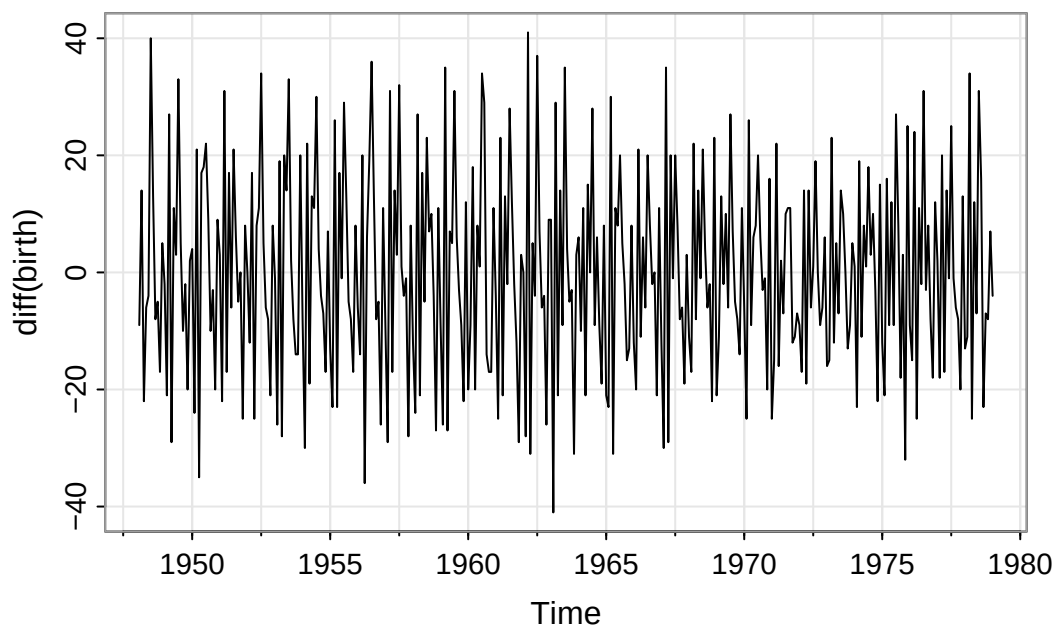
a)

We start our model process by investigating the time series plot:

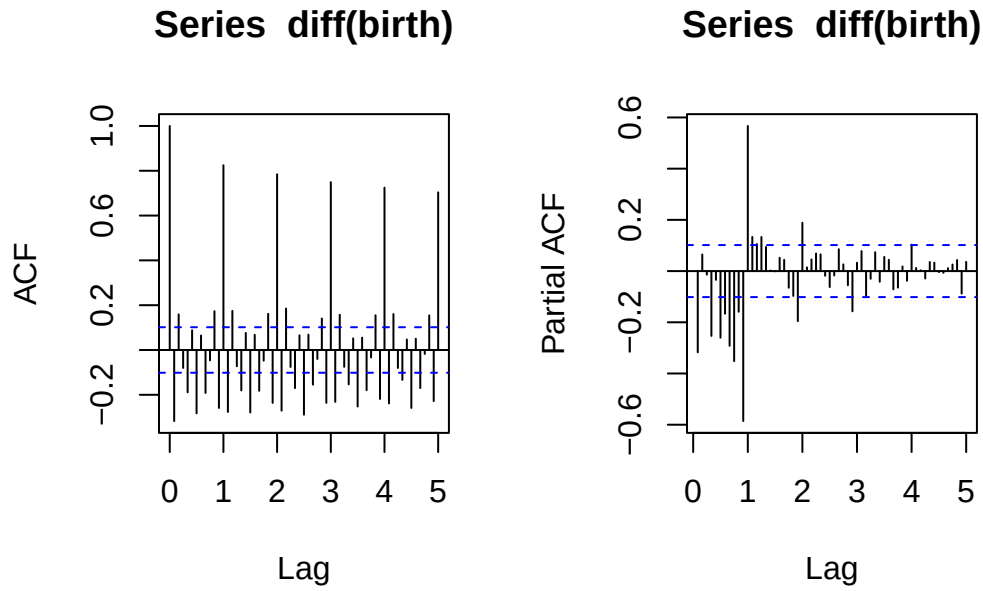


This plot looks a lot like a random walk process. Variance appears constant so no issues there. It looks like there are seasonal effects.

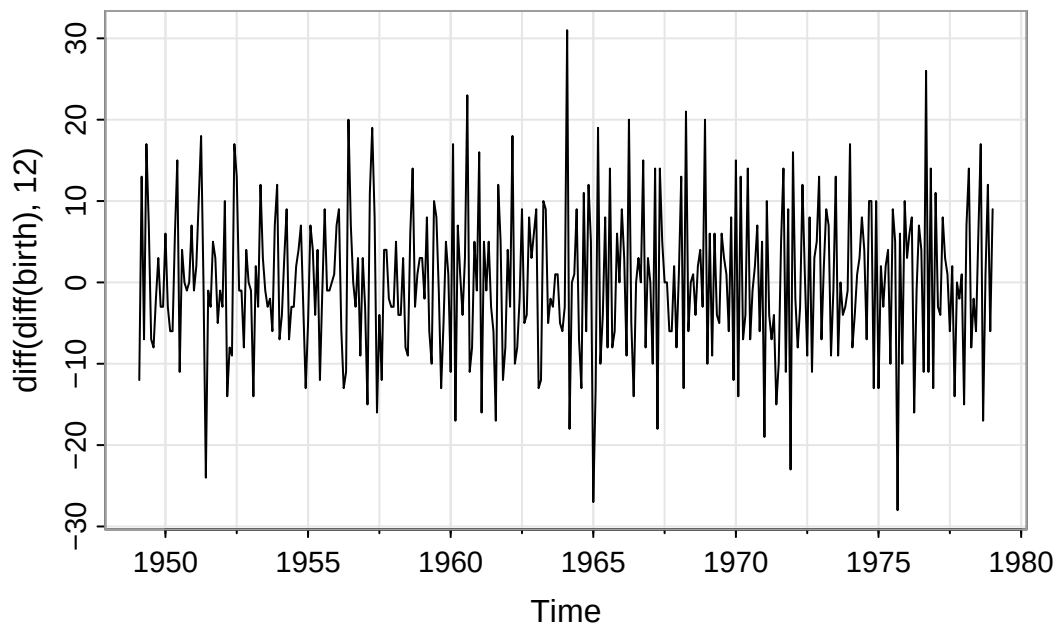
I first try a difference to deal with the random walk process.



This looks a lot closer to stationarity. To assess seasonality and choose p, q and P, Q we look at the ACF and PACF of the differenced data.

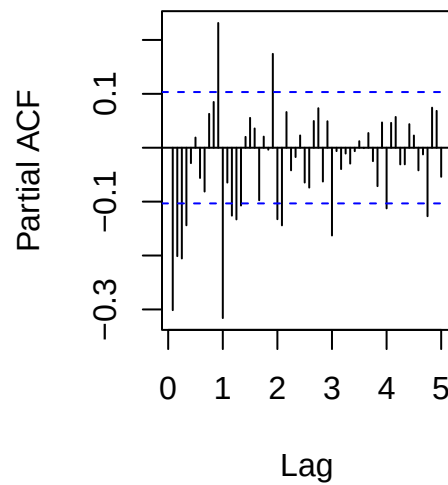
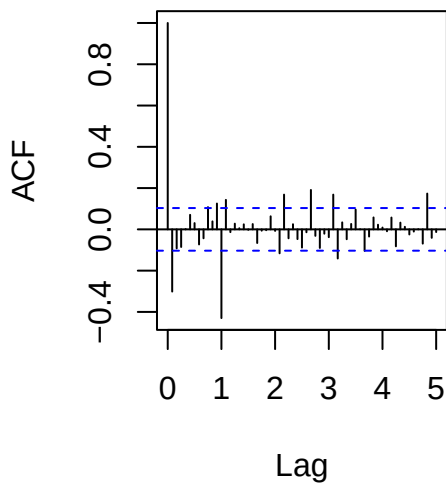


We see significant correlation between values a year apart. To go after this we different again using x_{t-12} :



Series $\text{diff}(\text{diff}(\text{birth}), 12)$

Series $\text{diff}(\text{diff}(\text{birth}), 12)$

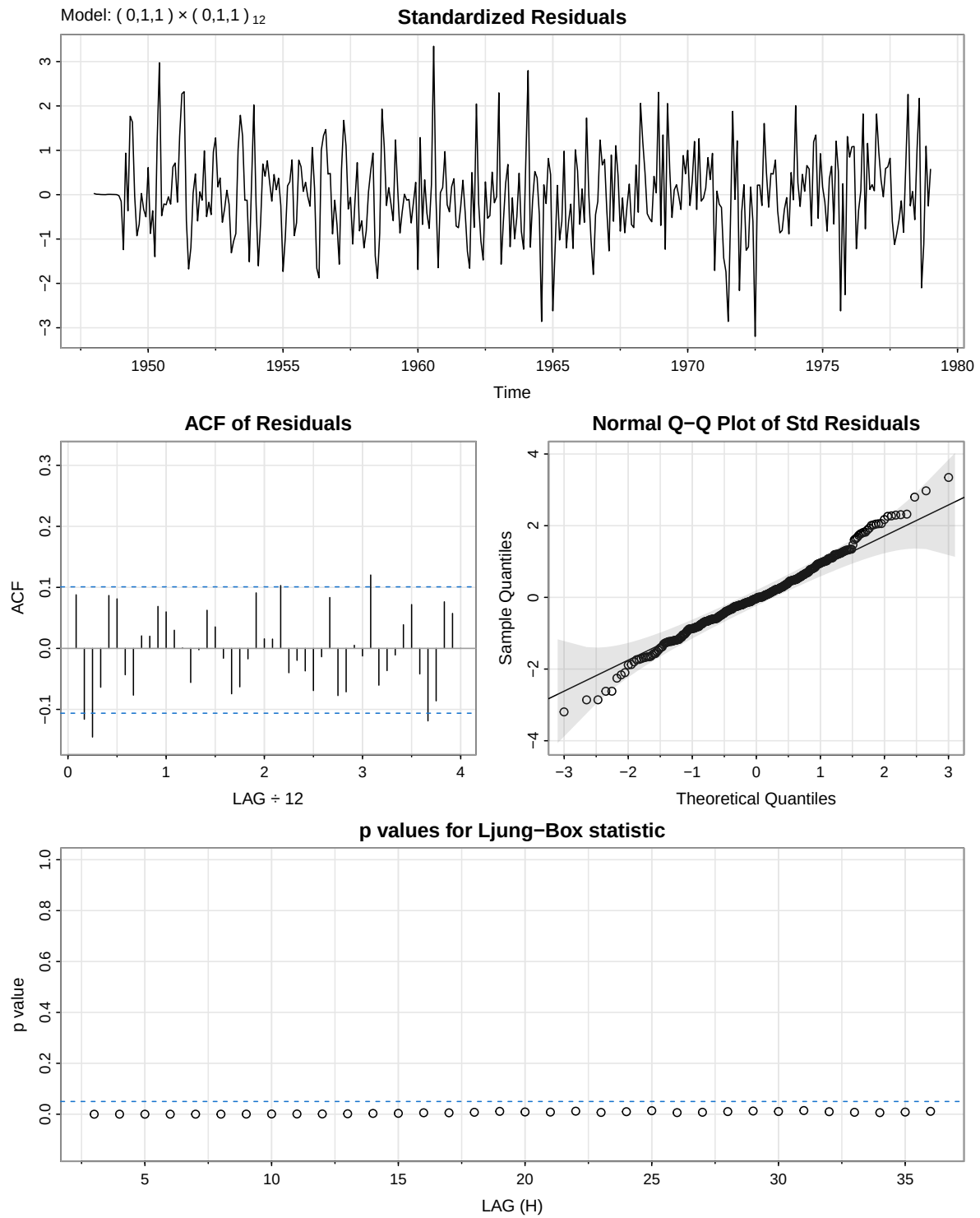


Now consulting the 5.1 table, we see that the ACF cuts off after the first 12 month lag and the PACF seems to tail off in 12 month intervals.

This would seem to indicate a $P = 0, Q = 1$ for the seasonal component of the $ARIMA(p, d, q) \times (P, D, Q)_s$.

σ^2 estimated as 47.40199 on 358 degrees of freedom

AIC = 6.74599 AICc = 6.746084 BIC = 6.778375

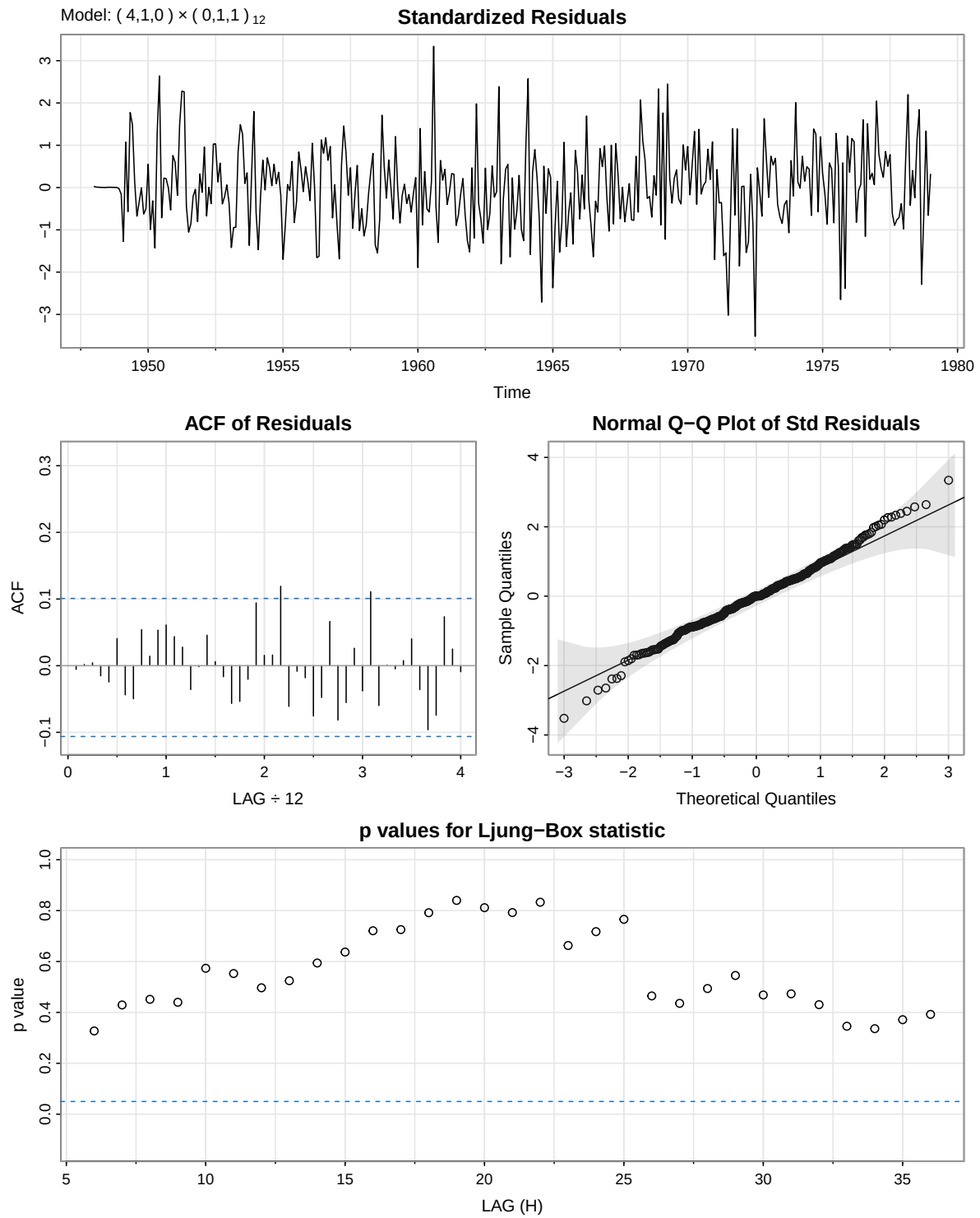


The $ARIMA(0,1,1) \times (0,1,1)_{12}$ model's biggest weakness is that there is still autocorrelation

Because of this I consider the other model $ARIMA(4, 1, 0) \times (0, 1, 1)_{12}$:

Coefficients:				
	Estimate	SE	t.value	p.value
ar1	-0.3875	0.0529	-7.3207	0.000
ar2	-0.2877	0.0550	-5.2315	0.000
ar3	-0.2502	0.0544	-4.6018	0.000
ar4	-0.1747	0.0528	-3.3070	0.001
sma1	-0.8060	0.0431	-18.6959	0.000

AIC = 6.715188 AICc = 6.715659 BIC = 6.779956



This model looks a lot better. Variance looks constant, the ACF plot and Ljung-Box statistic

plots support the assumption of white noise errors and residuals look normal in the qq-plot.

Despite the diagnostics looking better, I am still concerned about the complexity of using $p = 4$. So I also try $ARMA(1, 1, 1) \times (0, 1, 1)_{12}$

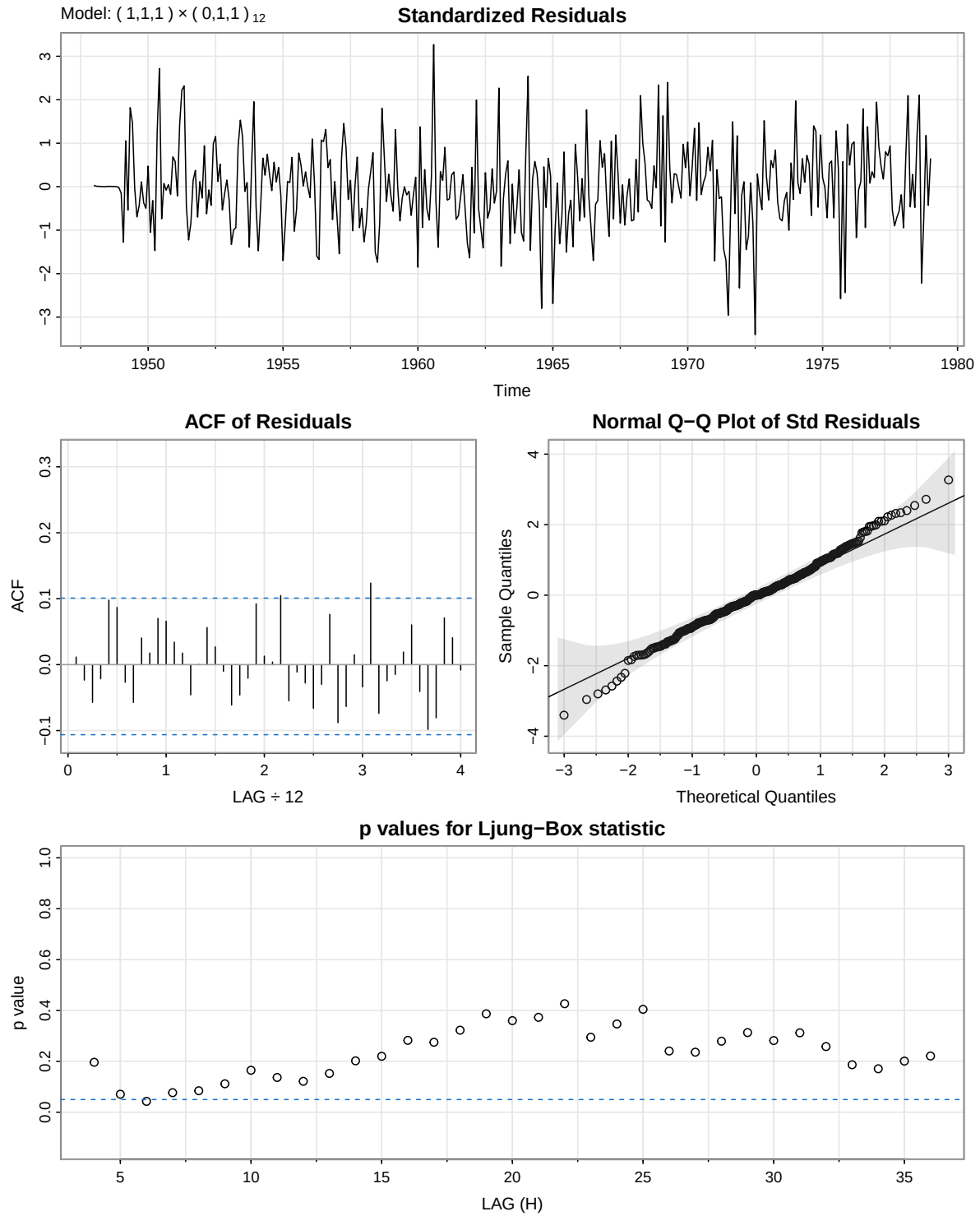
```
initial    value   2.218186  
iter      2 value   2.032584  
iter      3 value   1.982464  
iter      4 value   1.975643  
iter      5 value   1.971721  
iter      6 value   1.967284  
iter      7 value   1.963840  
iter      8 value   1.961106  
iter      9 value   1.960849  
iter     10 value   1.960692  
iter     11 value   1.960683  
iter     12 value   1.960675  
iter     13 value   1.960672  
iter     13 value   1.960672  
final    value   1.960672  
converged  
  
initial    value   1.940459  
iter      2 value   1.934425  
iter      3 value   1.932752  
iter      4 value   1.931750  
iter      5 value   1.931074  
iter      6 value   1.930882  
iter      7 value   1.930860  
iter      8 value   1.930859  
iter      8 value   1.930859  
final    value   1.930859  
converged
```

Coefficients:

	Estimate	SE	t.value	p.value
ar1	0.3038	0.0865	3.5104	5e-04
ma1	-0.7006	0.0604	-11.5984	0e+00
sma1	-0.8000	0.0441	-18.1302	0e+00

σ^2 estimated as 45.91462 on 357 degrees of freedom

AIC = 6.721818 AIC_c = 6.722006 BIC = 6.764997



The models are comparable except for the p-value for Ljung-Box statistic plot, this model

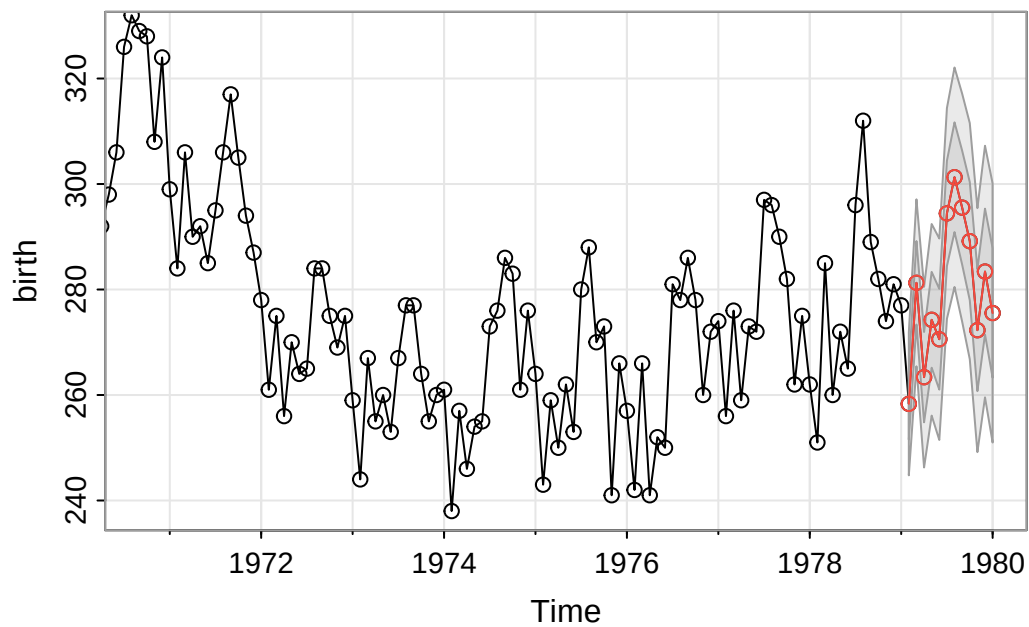
doesn't reject the null hypothesis around lag 7. But generally, the residuals are acceptable.

To make a final decision I compare the AIC/BIC of the two models:

Model	AIC	BIC
$ARIMA(0, 1, 4) \times (0, 1, 1)_{12}$	6.715	6.778
$ARIMA(1, 1, 1) \times (0, 1, 1)_{12}$	6.722	6.765

Using AIC we would select the more complex model, using BIC we would select the simpler model. I choose $ARIMA(1, 1, 1) \times (0, 1, 1)_{12}$ due to it being a simpler model.

The 12 month forecast is below:



\$pred

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug
1979	258.3256	281.2730	263.3864	274.2558	270.5740	294.4331	301.2872	
1980	275.5399							
	Sep	Oct	Nov	Dec				
1979	295.5201	289.1691	272.2940	283.4030				
1980								

\$se

	Jan	Feb	Mar	Apr	May	Jun	Jul
--	-----	-----	-----	-----	-----	-----	-----

```

1979          6.776034  7.913388  8.562644  9.080446  9.546734  9.984553
1980 12.274923
          Aug          Sep          Oct          Nov          Dec
1979 10.402012 10.802780 11.189036 11.562347 11.923961
1980

```

b)

The Facebook Prophet model has three components, trend, seasonal and holiday. It is like a generalized additive model, and its trying to fit the curve of data as well as possible (which causes it to lose statistical inference compared to ARIMA). The trend component can be either non-linear or piece wise linear. The seasonal component is a Fourier series which requires a choice of N, N = 3 is noted as good for weekly seasonality and N = 7 for yearly. The holiday component is a set of dummy variables corresponding to holidays specified by the analyst.

c)

Loading required package: Rcpp

Loading required package: rlang

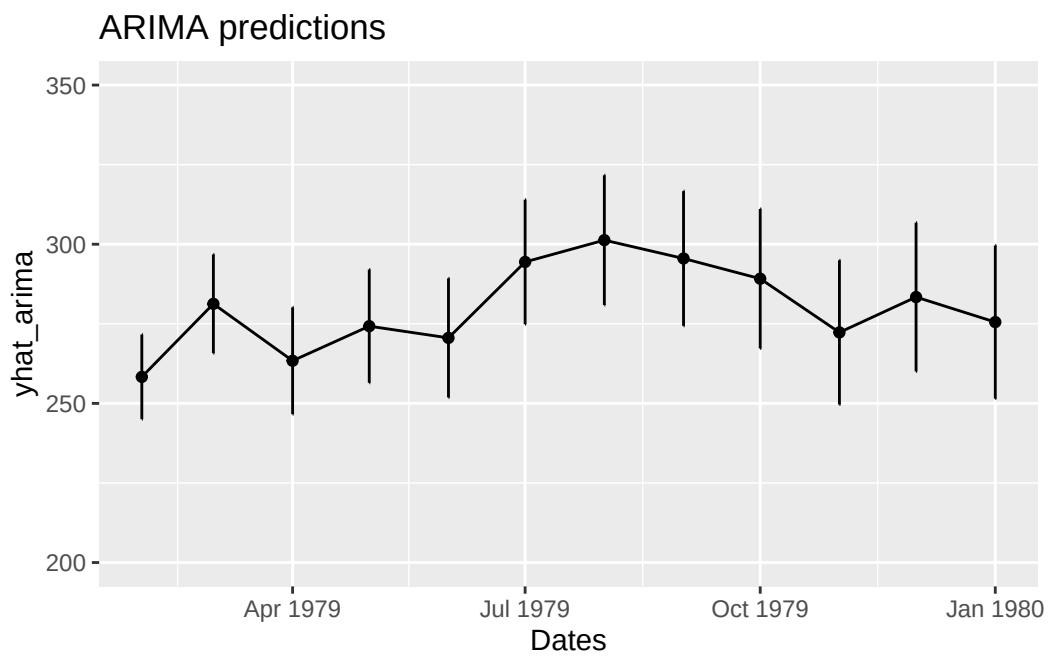
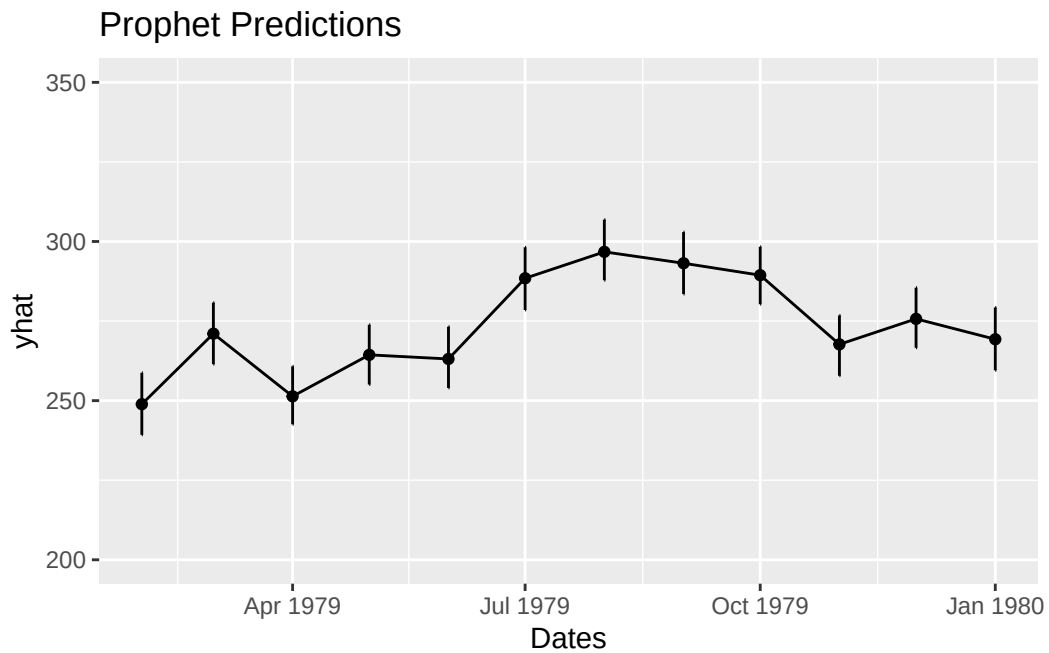
```

-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.1.4      v readr      2.1.5
v forcats    1.0.0      v stringr    1.5.1
v ggplot2    3.5.2      v tibble     3.2.1
v lubridate  1.9.4      v tidyr      1.3.1
v purrr      1.0.4
-- Conflicts ----- tidyverse_conflicts() --
x purrr::%@%()          masks rlang::%@%()
x dplyr::filter()       masks stats::filter()
x dplyr::first()        masks xts::first()
x purrr::flatten()      masks rlang::flatten()
x purrr::flatten_chr() masks rlang::flatten_chr()
x purrr::flatten_dbl() masks rlang::flatten_dbl()
x purrr::flatten_int() masks rlang::flatten_int()
x purrr::flatten_lgl() masks rlang::flatten_lgl()
x purrr::flatten_raw() masks rlang::flatten_raw()
x purrr::invoke()       masks rlang::invoke()
x dplyr::lag()          masks stats::lag()
x dplyr::last()         masks xts::last()
x purrr::splice()       masks rlang::splice()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become

```

Disabling weekly seasonality. Run prophet with `weekly.seasonality=TRUE` to override this.

Disabling daily seasonality. Run prophet with `daily.seasonality=TRUE` to override this.



We can see that the general trend is similar, but prophet has smaller prediction intervals interestingly enough. The ARIMA model predicts higher values at the starting date when compared to the prophet model.